

RESEARCH ARTICLE | JULY 02 2024

## Computation of three-dimensional incompressible flows using high-order weighted essentially non-oscillatory finite-difference lattice Boltzmann method

Kazem Hejranfar  ; Mohammad Abotalebi 



*Physics of Fluids* 36, 077107 (2024)

<https://doi.org/10.1063/5.0212459>



### Articles You May Be Interested In

A high-order generalized differential quadrature method with lattice Boltzmann flux solver for simulating incompressible flows

*Physics of Fluids* (April 2023)

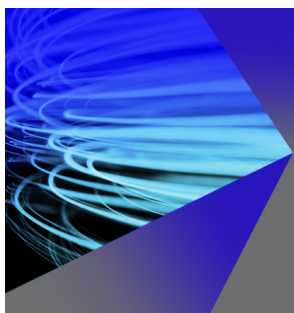
A cost-efficient third-order upwind compact scheme with adjustable dissipation for entropically damped artificial compressibility model-based transient incompressible flows

*Physics of Fluids* (March 2025)

Accelerate the numerical convergence of incompressible flows: Novel preconditioned characteristic boundary conditions

*Physics of Fluids* (February 2024)

28 March 2026 05:22:37



## AIP Advances

### Why Publish With Us?



**21DAYS**  
average time  
to 1st decision



**OVER 4 MILLION**  
views in the last year



**INCLUSIVE**  
scope

[Learn More](#)

 AIP  
Publishing

# Computation of three-dimensional incompressible flows using high-order weighted essentially non-oscillatory finite-difference lattice Boltzmann method

Cite as: Phys. Fluids **36**, 077107 (2024); doi: 10.1063/5.0212459

Submitted: 4 April 2024 · Accepted: 12 June 2024 ·

Published Online: 2 July 2024



View Online



Export Citation



CrossMark

Kazem Hejranfar<sup>a)</sup>  and Mohammad Abotalebi 

## AFFILIATIONS

Aerospace Engineering Department, Sharif University of Technology, Tehran, Iran

<sup>a)</sup> Author to whom correspondence should be addressed: [khejran@sharif.edu](mailto:khejran@sharif.edu)

## ABSTRACT

In the present work, an accurate and robust solution methodology based on the high-order weighted essentially non-oscillatory (WENO) finite-difference lattice Boltzmann method (LBM) in the three-dimensional generalized curvilinear coordinates is presented and applied for simulating the three-dimensional incompressible flows over complicated configurations with curved boundaries. Here, the incompressible form of the lattice Boltzmann equation in three dimensions is considered and the discretization of the spatial derivative terms is performed with the fifth-order WENO finite-difference method and the temporal derivative term is discretized with the fourth-order Runge–Kutta scheme to ensure the accuracy and stability of the solution method for both the steady and unsteady problems. The three-dimensional lattice Boltzmann equation applied here is based on a nineteen discrete velocity model for transforming the microscopic properties to the macroscopic ones. To assess the accuracy and robustness of the present three-dimensional high-order finite-difference LBM solver, different incompressible flow benchmarks and practical test cases are studied that are the cavity flow, the Beltrami flow, the flow in the curved ducts of rectangular cross sections, and the flow over a sphere for different flow conditions. The decay of the homogeneous isotropic turbulence is also computed to examine the suitability of the present solution method to be applied as the direct numerical simulation of turbulent flows. It is demonstrated that the solution methodology presented based on the high-order WENO finite-difference LBM in the three-dimensional generalized curvilinear coordinate can be used for accurately and effectively computing the three-dimensional practical incompressible flow problems.

Published under an exclusive license by AIP Publishing. <https://doi.org/10.1063/5.0212459>

## I. INTRODUCTION

In the past three decades, the benefits of the lattice Boltzmann method (LBM), namely, the simplicity in numerical implementation, the existence of linear convection terms, and an inherently parallelizable algorithm, have introduced this solution method as an attractive alternative to the conventional Navier–Stokes flow solvers. The lattice Boltzmann (LB) equation is a simple linear hyperbolic equation based on the statistical mechanics and the kinetic theory, and this approach relates the mesoscopic scale of the flow to the macroscopic one by using the distribution functions defined in the discretized microscopic velocity space. The independent sets of the microscopic velocities and also the corresponding distribution functions in the LBM provide the capability of efficiently parallelizing its solution algorithm. On the other hand, the second-order accuracy in time and space, the

convergence and stability issues at high Reynolds number flows, and the limit of using the uniform Cartesian grid systems are the main restrictions of the standard LBM with the Bhatnagar–Gross–Krook (BGK) approximation.

Some efforts have been made in the literature to improve the accuracy and applicability of the standard LBM in precisely simulating the fluid flows over the practical geometries. The study of the flow in the curved domains has made by Nannelli and Succi,<sup>1</sup> where they used a finite-volume scheme for discretizing conventional terms in the LB equation. After that, some well-known incompressible flow problems have been solved by Mie and Shyy<sup>2</sup> by transporting the LB equation to the curvilinear coordinates and using the second-order upwind finite-difference scheme for discretizing the first-order distribution function derivatives. He and Luo<sup>3</sup> expressed that the coupled time and space

discretization can be performed in a decouple manner and the high-order discretization can be implemented on the LB equation. The discretization procedures like finite volume,<sup>4–11</sup> finite difference,<sup>12–22</sup> finite/spectral element,<sup>23–25</sup> Chebyshev collocation spectral,<sup>26,27</sup> and spectral difference<sup>28,29</sup> methods have been used for improving the spatial and temporal accuracy of LBM. For increasing the stability limit of the LBM, the BGK model can be substituted with more efficient collision operators like the multi-relaxation-time (MRT),<sup>30,31</sup> cascaded,<sup>32,33</sup> entropy,<sup>34,35</sup> and cumulant<sup>36,37</sup> LBMs.

The ease of implementation in finite-difference methods and available high-order schemes made these methods more applicable than the other numerical schemes to be used in the framework of the LBM. In addition, the compatibility of the finite-difference method with the curved boundary by using the coordinate transformation to the generalized curvilinear coordinates introduces this procedure as a suitable choice for studying the flows over the complicated geometries with curved boundaries. The high-order central compact finite-difference method can be used for discretizing the LB equation; however, it needs a suitable filtering procedure to ensure the solution to be stable.<sup>16,17</sup> One can apply the upwind compact finite-difference method for discretizing the LB equation that does not need a filtering scheme for the stability considerations.<sup>18</sup> An alternative choice is to use the high-order weighted essentially non-oscillatory (WENO) finite-difference scheme for the discretization of the spatial derivative terms in the LB equation, and this solution method does not also require a filtering procedure.<sup>21,22</sup> Although the WENO scheme has originally been developed and applied in the literature for computing the compressible flows (for example, see Refs. 38–54) owing to its stability and robustness in dealing with the strong shocks/discontinuities, this high-order numerical scheme is also a proper choice for accurately simulating the incompressible flow problems (for example, see Refs. 55–58). Note that non-linear flow features occur in the flow field at high Reynolds number flows even for the incompressible flows. In addition, grid nonuniformity and boundary conditions may be the other sources of the numerical instability when computing the incompressible flows and these issues can be properly resolved with applying the WENO scheme.

The main objective of this paper is to extend and apply the high-order WENO finite-difference LBM (WENOLBM) in the three-dimensional generalized curvilinear coordinates for effectively computing the three-dimensional incompressible flows over practical geometries. The solution method in three dimensions is nearly similar to that applied in two dimensions as given in Refs. 21 and 22. Herein, the nineteen microscopic velocity model (D3Q19) is used for the three-dimensional LB equation and the fifth-order WENO finite-difference method is implemented for the spatial discretization and the fourth-order Runge–Kutta (RK4) method for the time integration in the LB equation. The accuracy and robustness of this high-order incompressible LB-based solver are examined by solving different three-dimensional incompressible flow problems.

The construction of the LB equation based on the three-dimensional incompressible model of the microscopic velocities (D3Q19) and its transformation to the three-dimensional generalized curvilinear coordinate are presented in Secs. II and III, respectively. After that, the spatial and temporal discretization of the LB equation is presented in Sec. IV based on the fifth-order WENO finite-difference and the fourth-order Runge–Kutta methods, respectively. Section V

corresponds to the implementation of the boundary conditions based on the microscopic and macroscopic properties. Results obtained by using the current solver for different incompressible flow problems are presented in Sec. VI and compared with the available numerical and experimental references to assess the accuracy and capability of the high-order WENO finite-difference LBM for analyzing practical and realistic incompressible flow problems.

## II. GOVERNING EQUATIONS

Recently, the robustness and accuracy of the WENO finite-difference LBM (WENOLBM) have been studied in the two-dimensional generalized curvilinear coordinates for solving the two-dimensional incompressible flows.<sup>21,22</sup> The main objective of the present study is to extend this solution methodology to three dimensions for effectively solving the three-dimensional incompressible flows over more realistic and practical geometries.

The LB equation based on the particle distribution function  $f(\mathbf{x}, \mathbf{e}, t)$  by using the BGK model<sup>59</sup> as the collision term can be presented as follows:<sup>60</sup>

$$\frac{\partial f_\alpha}{\partial t} + \mathbf{e}_\alpha \cdot \nabla f_\alpha = -\frac{1}{\tau} (f_\alpha - f_\alpha^{eq}), \quad \alpha = 0, \dots, 18, \quad (1)$$

where the right-hand side of this equation represents the relaxation of the particle populations toward an equilibrium distribution function as  $f_\alpha^{eq}$  at a rate determined by the relaxation time  $\tau$ , and the left-hand side shows the propagation of the distribution function by the particle velocity vector  $\mathbf{e}_\alpha$ , in which the subscript  $\alpha$  denotes the direction of the particle speed. The microscopic velocities  $\mathbf{e}_\alpha$  for the D3Q19 discrete Boltzmann model are given in Table I, and they are shown schematically in Fig. 1.

Based on the model presented by He and Luo,<sup>61</sup> through the Chapman–Enskog expansion, the LB equation satisfies the incompressible Navier–Stokes equations and the corresponding equilibrium distribution function is determined as follows:

$$f_\alpha^{eq} = \omega_\alpha \left( p + p_0 \left( 3 \frac{[\mathbf{e}_\alpha \cdot \mathbf{u}]}{c^2} + \frac{9 [\mathbf{e}_\alpha \cdot \mathbf{u}]^2}{2 c^4} - \frac{3 |\mathbf{u}|^2}{2 c^2} \right) \right), \quad (2)$$

where the vector  $\mathbf{u} = (u_x, u_y, u_z)$  and the scalar  $p$  are the macroscopic velocity vector and the pressure, respectively; the parameter  $c$  is the lattice (or particle) velocity; and the weight coefficient  $\omega_\alpha$  is introduced in Table I. The corresponding macroscopic properties are calculated by using the following relations:

$$p = \sum_{\alpha=0}^{18} f_\alpha, \quad p_0 \mathbf{u} = \sum_{\alpha=0}^{18} \mathbf{e}_\alpha f_\alpha, \quad (3)$$

**TABLE I.** Properties of the most popular velocity compatible with sound speed  $c_s = c/\sqrt{3}$ .

| $\mathbf{e}_\alpha = (e_{\alpha x}, e_{\alpha y}, e_{\alpha z})$ | $\omega_\alpha$ | $\alpha$ |
|------------------------------------------------------------------|-----------------|----------|
| $c(0, 0, 0)$                                                     | 1/3             | 0        |
| $c(\pm 1, 0, 0), c(0, \pm 1, 0), c(0, 0, \pm 1)$                 | 1/18            | 1–6      |
| $c(\pm 1, \pm 1, 0), c(\pm 1, 0, \pm 1), c(0, \pm 1, \pm 1)$     | 1/36            | 7–18     |

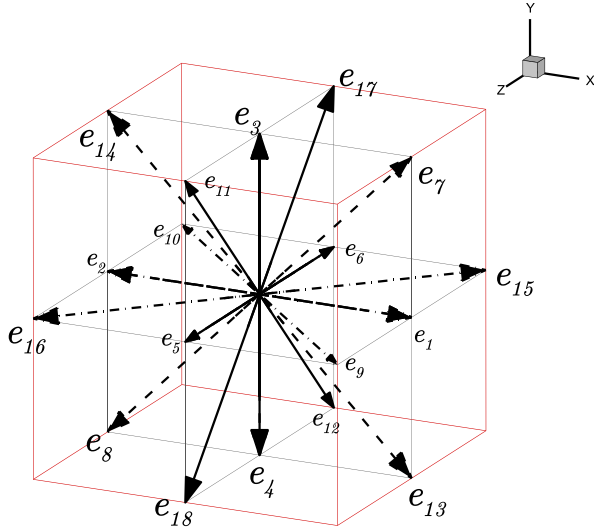


FIG. 1. Cubic D3Q19 lattice model and the microscopic velocities.

where  $p_0 = c_s^2 \rho_0$  is the pressure corresponding to the constant density  $\rho_0$  of the fluid and  $c_s = c/\sqrt{3}$  is the sound speed that relates the kinematic viscosity  $\nu$  of the fluid to the relaxation time  $\tau$  through the following relation:

$$\tau = \frac{\nu}{c_s^2}. \quad (4)$$

Here, the particle velocity  $c$  is set to be unity, and therefore, the velocity field is normalized with this parameter and  $c_s = 1/\sqrt{3}$  in the lattice units. The kinematic viscosity  $\nu$  is calculated by fixing the Reynolds number  $Re = u_{ref}L/\nu$ , which is defined based on the reference velocity  $u_{ref}$  and the reference length  $L$ .

### III. TRANSFORMATION TO GENERALIZED CURVILINEAR COORDINATE

In this section, by considering  $x_i = (x, y, z)$  and  $\xi^i = (\xi, \eta, \zeta)$  as the components of physical and computational domains, respectively, and their relations as follows:

$$\begin{bmatrix} \xi_x & \eta_x & \zeta_x \\ \xi_y & \eta_y & \zeta_y \\ \xi_z & \eta_z & \zeta_z \end{bmatrix} = \begin{bmatrix} x_\xi & x_\eta & x_\zeta \\ y_\xi & y_\eta & y_\zeta \\ z_\xi & z_\eta & z_\zeta \end{bmatrix}^{-1}, \quad (5)$$

and the corresponded transformation Jacobian as

$$J = \det \left[ \frac{\partial x_i}{\partial \xi^j} \right] = \begin{vmatrix} x_\xi & y_\xi & z_\xi \\ x_\eta & y_\eta & z_\eta \\ x_\zeta & y_\zeta & z_\zeta \end{vmatrix}, \quad (6)$$

and using the chain rule, the LB equation (1) is transformed to the three-dimensional generalized curvilinear coordinates. Since the curvature effects only appear in the spatial derivative terms, only the convection term is transformed as follows:

$$\begin{aligned} e_\alpha \cdot \nabla f_\alpha &= e_{\alpha x} \frac{\partial f_\alpha}{\partial x} + e_{\alpha y} \frac{\partial f_\alpha}{\partial y} + e_{\alpha z} \frac{\partial f_\alpha}{\partial z} \\ &= e_{\alpha x} \left( \frac{\partial \xi}{\partial x} \frac{\partial f_\alpha}{\partial \xi} + \frac{\partial \eta}{\partial x} \frac{\partial f_\alpha}{\partial \eta} + \frac{\partial \zeta}{\partial x} \frac{\partial f_\alpha}{\partial \zeta} \right) \\ &\quad + e_{\alpha y} \left( \frac{\partial \xi}{\partial y} \frac{\partial f_\alpha}{\partial \xi} + \frac{\partial \eta}{\partial y} \frac{\partial f_\alpha}{\partial \eta} + \frac{\partial \zeta}{\partial y} \frac{\partial f_\alpha}{\partial \zeta} \right) \\ &\quad + e_{\alpha z} \left( \frac{\partial \xi}{\partial z} \frac{\partial f_\alpha}{\partial \xi} + \frac{\partial \eta}{\partial z} \frac{\partial f_\alpha}{\partial \eta} + \frac{\partial \zeta}{\partial z} \frac{\partial f_\alpha}{\partial \zeta} \right) \\ &= (e_{\alpha x} \xi_x + e_{\alpha y} \xi_y + e_{\alpha z} \xi_z) \frac{\partial f_\alpha}{\partial \xi} + (e_{\alpha x} \eta_x + e_{\alpha y} \eta_y + e_{\alpha z} \eta_z) \frac{\partial f_\alpha}{\partial \eta} \\ &\quad + (e_{\alpha x} \zeta_x + e_{\alpha y} \zeta_y + e_{\alpha z} \zeta_z) \frac{\partial f_\alpha}{\partial \zeta}. \end{aligned} \quad (7)$$

For simplicity, using the definition of the microscopic contravariant velocity vector as

$$\begin{aligned} \tilde{e}_\alpha &= (\tilde{e}_{\alpha \xi}, \tilde{e}_{\alpha \eta}, \tilde{e}_{\alpha \zeta}) \\ &= (e_{\alpha x} \xi_x + e_{\alpha y} \xi_y + e_{\alpha z} \xi_z, e_{\alpha x} \eta_x + e_{\alpha y} \eta_y + e_{\alpha z} \eta_z, e_{\alpha x} \zeta_x + e_{\alpha y} \zeta_y + e_{\alpha z} \zeta_z). \end{aligned} \quad (8)$$

Equation (1) can be represented in the generalized curvilinear coordinates as follows:

$$\frac{\partial f_\alpha}{\partial t} + \tilde{e}_{\alpha \xi} \frac{\partial f_\alpha}{\partial \xi} + \tilde{e}_{\alpha \eta} \frac{\partial f_\alpha}{\partial \eta} + \tilde{e}_{\alpha \zeta} \frac{\partial f_\alpha}{\partial \zeta} = -\frac{1}{\tau} (f_\alpha - f_\alpha^{eq}), \quad \alpha = 0, \dots, 18. \quad (9)$$

Note that the formulation used here is based on the nonconservative form of the governing equations. Alternatively, one can use the conservative form of the governing equations and these two formulations give the same solution for the incompressible flows.

### IV. DISCRETIZATION PROCEDURE

In this section, the fifth-order WENO method for the discretization of the spatial derivative terms and the fourth-order Runge-Kutta scheme for the discretization of the temporal derivative term in the three-dimensional LB equation in the generalized curvilinear coordinates are described. The main advantage of the high-order WENO finite-difference scheme over the other high-order finite-difference methods like the central compact scheme is that it has the non-oscillating property providing a suitable and robust flow solver without the need for any filtering procedure to make the solution method to be stable.

The three-dimensional form of the LB equation in the generalized curvilinear coordinates (9) can be rewritten as follows:

$$\begin{aligned} \frac{\partial f_\alpha}{\partial t} &= R(f_\alpha), \\ R(f_\alpha) &= -\left( \tilde{e}_{\alpha \xi} \frac{\partial f_\alpha}{\partial \xi} + \tilde{e}_{\alpha \eta} \frac{\partial f_\alpha}{\partial \eta} + \tilde{e}_{\alpha \zeta} \frac{\partial f_\alpha}{\partial \zeta} \right) - \frac{1}{\tau} (f_\alpha - f_\alpha^{eq}). \end{aligned} \quad (10)$$

Here, the modified explicit fourth-order Runge-Kutta (RK4) scheme<sup>62</sup> is used for the time integration of Eq. (10) in the following four steps:

$$\begin{aligned} f_\alpha^0 &= f_\alpha^t, \\ f_\alpha^k &= f_\alpha^0 + \ell_k \Delta t R(f_\alpha^{k-1}), \quad k = 1, 2, 3, 4, \quad \ell_k = \frac{1}{4}, \frac{1}{3}, \frac{1}{2}, 1, \\ f_\alpha^{t+\Delta t} &= f_\alpha^4. \end{aligned} \quad (11)$$



To check the accuracy and performance of this explicit time integration scheme, another explicit time integration scheme, namely, the low-storage 10-stage 4th-order strong stability-preserving Runge-Kutta (SSP-RK4) method<sup>63</sup> is also applied. This time integration scheme has been combined with the WENO finite-difference method for solving the ideal magnetohydrodynamic (MHD) equations.<sup>47</sup> The implementation of this time integration scheme to Eq. (10) is expressed as follows:

$$\begin{aligned} f_x^{1,1} &= f_x^n, \quad f_x^{2,1} = f_x^n, \\ f_x^{1,k+1} &= f_x^{1,k} + \frac{\Delta t}{6} R(f_x^{1,k}), \quad k = 1, 2, 3, 4, 5, \\ f_x^{2,2} &= \frac{1}{25} f_x^{2,1} + \frac{9}{25} f_x^{1,k}, \quad k = 5, \\ f_x^{1,k+1} &= 15 f_x^{2,2} - 5 f_x^{1,k}, \quad k = 5, \\ f_x^{1,k+1} &= f_x^{1,k} + \frac{\Delta t}{6} R(f_x^{1,k}), \quad k = 6, 7, 8, \\ f_x^{1,k+1} &= f_x^{2,2} + \frac{3}{5} f_x^{1,k} + \frac{\Delta t}{10} R(f_x^{1,k}), \quad k = 9. \end{aligned} \quad (12)$$

These two explicit time integration schemes implemented in the WENOLBM are compared with each other in terms of the accuracy and performance by computing the two-dimensional transient incompressible flow inside the cavity and also the three-dimensional Beltrami flow (see the numerical results section).

Now, the discretization of the spatial derivative terms appeared in Eq. (9) by applying the high-order WENO finite-difference scheme<sup>21,22,38–40</sup> is described. The first derivative of the distribution function in the computational plane in each direction, like  $\xi$ , i.e.,  $\partial f_x / \partial \xi$ , can be presented in term of the cell-midpoint values of  $f_x$  as follows:

$$\frac{\partial f_{x,i}}{\partial \xi} = \frac{1}{\Delta \xi} (f_{x,i+\frac{1}{2}} - f_{x,i-\frac{1}{2}}). \quad (13)$$

As shown in Fig. 2, the fifth-order WENO finite-difference scheme uses a five-points stencil for estimating the cell-midpoint values of  $f_x$  that is divided into the three-point stencils as the candidate stencils. In the smooth region, these three stencils are combined linearly and construct the main five-point stencil that is an optimal situation which the scheme achieves the fifth-order accuracy. In the non-smooth regions, a

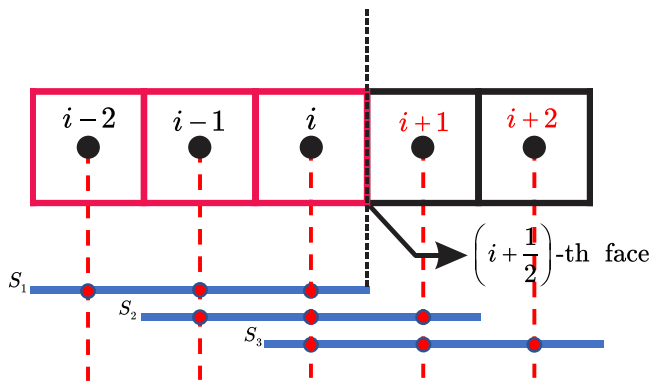


FIG. 2. Candidate 3-point stencils in the fifth-order WENO scheme.

non-linear combination of these three stencils must be used to reach the non-oscillating property of the method, the result is a variation of the order of accuracy between the three and five in the non-smooth regions. These three stencils for the positive contravariant microscopic velocities ( $\tilde{e}_{\alpha\xi} \geq 0$ ) can be given as follows:

$$\begin{aligned} \hat{f}_{x,i+\frac{1}{2}}^1 &= \frac{1}{3} f_{x,i-2} - \frac{7}{6} f_{x,i-1} + \frac{11}{6} f_{x,i}, \\ \hat{f}_{x,i+\frac{1}{2}}^2 &= -\frac{1}{6} f_{x,i-1} + \frac{5}{6} f_{x,i} + \frac{1}{3} f_{x,i+1}, \\ \hat{f}_{x,i+\frac{1}{2}}^3 &= \frac{1}{3} f_{x,i} + \frac{5}{6} f_{x,i+1} - \frac{1}{6} f_{x,i+2}. \end{aligned} \quad (14)$$

To achieve the non-oscillating property of the WENO finite-difference scheme, these three stencils must be combined with a non-linear weight to form the cell-midpoint value of  $f_x$

$$f_{x,i+\frac{1}{2}} = w_1 \hat{f}_{x,i+\frac{1}{2}}^1 + w_2 \hat{f}_{x,i+\frac{1}{2}}^2 + w_3 \hat{f}_{x,i+\frac{1}{2}}^3, \quad (15)$$

where  $w_k$  are the non-linear weight coefficients for optimizing the overall accuracy and they are expressed in the following relations:

$$w_k = \frac{\hat{w}_k}{\hat{w}_1 + \dots + \hat{w}_\ell}, \quad \hat{w}_k = \frac{\gamma_k}{(\varepsilon + \beta_k)^p}, \quad k = 1, \dots, \ell, \quad (16)$$

where  $\gamma_1 = 1/10$ ,  $\gamma_2 = 6/10$ ,  $\gamma_3 = 3/10$  are the optimal weight coefficients and the combination of the candidate stencils with these coefficients lead to the five-point stencil with the fifth-order accuracy capable only in the smooth region and it means that in the smooth region,  $w_k = \gamma_k$ . To avoid a zero denominator in the determination of the weight coefficients  $w_k$ , a constant parameter expressed by  $\varepsilon$  is entered in the corresponding relation in Eq. (16). Jiang and Shu<sup>39</sup> pointed out that the numerical calculations are not affected when this parameter has a value in the interval  $10^{-7} \leq \varepsilon \leq 10^{-5}$ . The parameter  $\ell$  is the number of the candidate stencils which  $\ell = 3$  for the fifth-order WENO finite-difference scheme and  $\beta_k$  is the smoothness indicator that is obtained from the following relation:

$$\beta_k = \sum_{j=1}^{\ell-1} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} (\Delta x)^{2j-1} \left( \frac{\partial \hat{f}_k(x)}{\partial x^j} \right)^2 dx, \quad (17)$$

and the corresponding smoothness indicators are determined as follows:

$$\begin{aligned} \beta_1 &= \frac{13}{12} (f_{x,i-2} - 2f_{x,i-1} + f_{x,i})^2 + \frac{1}{4} (f_{x,i-2} - 4f_{x,i-1} + 3f_{x,i})^2, \\ \beta_2 &= \frac{13}{12} (f_{x,i-1} - 2f_{x,i} + f_{x,i+1})^2 + \frac{1}{4} (f_{x,i-1} - f_{x,i+1})^2, \\ \beta_3 &= \frac{13}{12} (f_{x,i} - 2f_{x,i+1} + f_{x,i+2})^2 + \frac{1}{4} (3f_{x,i} - 4f_{x,i+1} + f_{x,i+2})^2. \end{aligned} \quad (18)$$

The parameter  $\tilde{p}$  in Eq. (16) is related to the dissipation error of the scheme in the non-smooth region and different values may be used for this parameter. Jiang and Shu<sup>39</sup> mentioned that for achieving the non-oscillatory property when  $\ell = 2, 3$ , the choice  $\tilde{p} = 2$  is proper and larger values of this parameter can reduce the accuracy of the scheme by adding more numerical dissipation to the solution. On the other hand, Lui *et al.*<sup>38</sup> pointed out that the necessity of being a non-oscillatory solution is  $\tilde{p} = \ell$ . Thus,  $\tilde{p} = 2$  is considered in the present study.

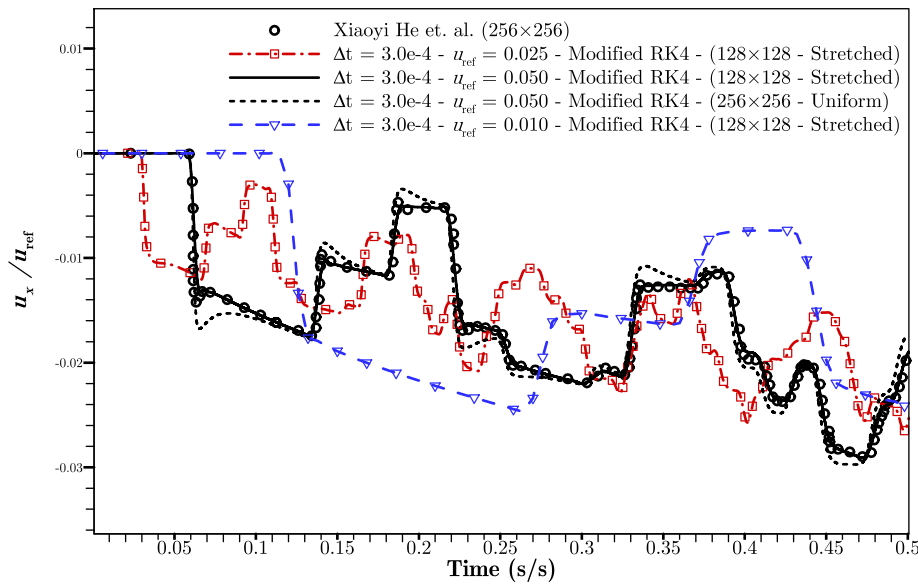


FIG. 3. Comparison of transient solution of the  $u$ -velocity variable at the center of the 2D cavity with  $Re = 1000$  for different values of reference velocity.

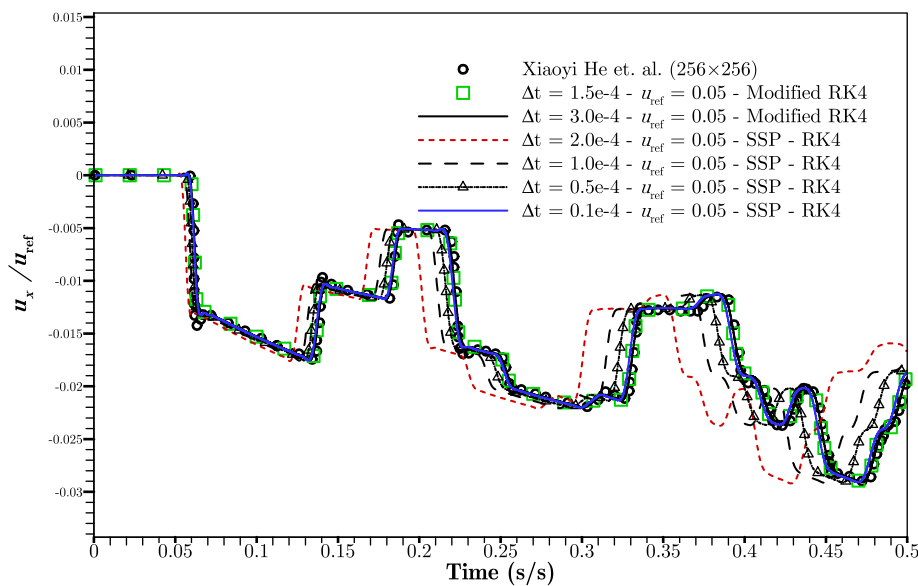


FIG. 4. Comparison of transient solution of the  $u$ -velocity variable at the center of the 2D cavity with  $Re = 1000$  using the SSP-RK4 and modified-RK4 schemes.

With the above formulation, the value of  $f_{x,i+1/2}$  can be calculated in the intervals of  $3 \leq i \leq I_{\max} - 2$  for the positive values of the contravariant microscopic velocity  $\tilde{e}_{x\zeta} \geq 0$  and in the interval of  $2 \leq i \leq I_{\max} - 3$  for  $\tilde{e}_{x\zeta} \leq 0$ . For calculating  $f_{x,i+1/2}$  at the near boundary

**TABLE II.** Comparison of the CPU time and stability threshold for the SSP-RK4 and modified-RK4 schemes for the 2D cavity problem with  $Re = 1000$ .

| Scheme       | $\Delta t/\tau$ | CPU time for steady-state solution (h) | CPU time for accurate transient solution (h) |
|--------------|-----------------|----------------------------------------|----------------------------------------------|
| SSP-RK4      | 5.0             | 70.23                                  | 12.20                                        |
| Modified-RK4 | 2.0             | 86.50                                  | 0.40                                         |

points, one-sided finite-difference relations should be used and a suitable choice is the use of the relations (14), which are more compatible with the main algorithm. These relations with the third-order accuracy are given below

$$\begin{cases} i = 1, I_{\max} - 2 : f_{x,i+\frac{1}{2}} = \frac{1}{3}f_{x,i} + \frac{5}{6}f_{x,i+1} - \frac{1}{6}f_{x,i+2}, \\ i = 2, I_{\max} - 1 : f_{x,i+\frac{1}{2}} = -\frac{1}{6}f_{x,i-1} + \frac{5}{6}f_{x,i} + \frac{1}{3}f_{x,i+1}. \end{cases} \quad (19)$$

Note that the relations given in this section are usable for the positive microscopic velocities  $\tilde{e}_{x\zeta} \geq 0$ . Similar relations for the negative microscopic velocities  $\tilde{e}_{x\zeta} \leq 0$  can be achieved by changing the stencils symmetrically regarding right or left cell boundaries. In addition, the distribution function relations at the left cell boundaries  $f_{x,i-\frac{1}{2}}$  can be

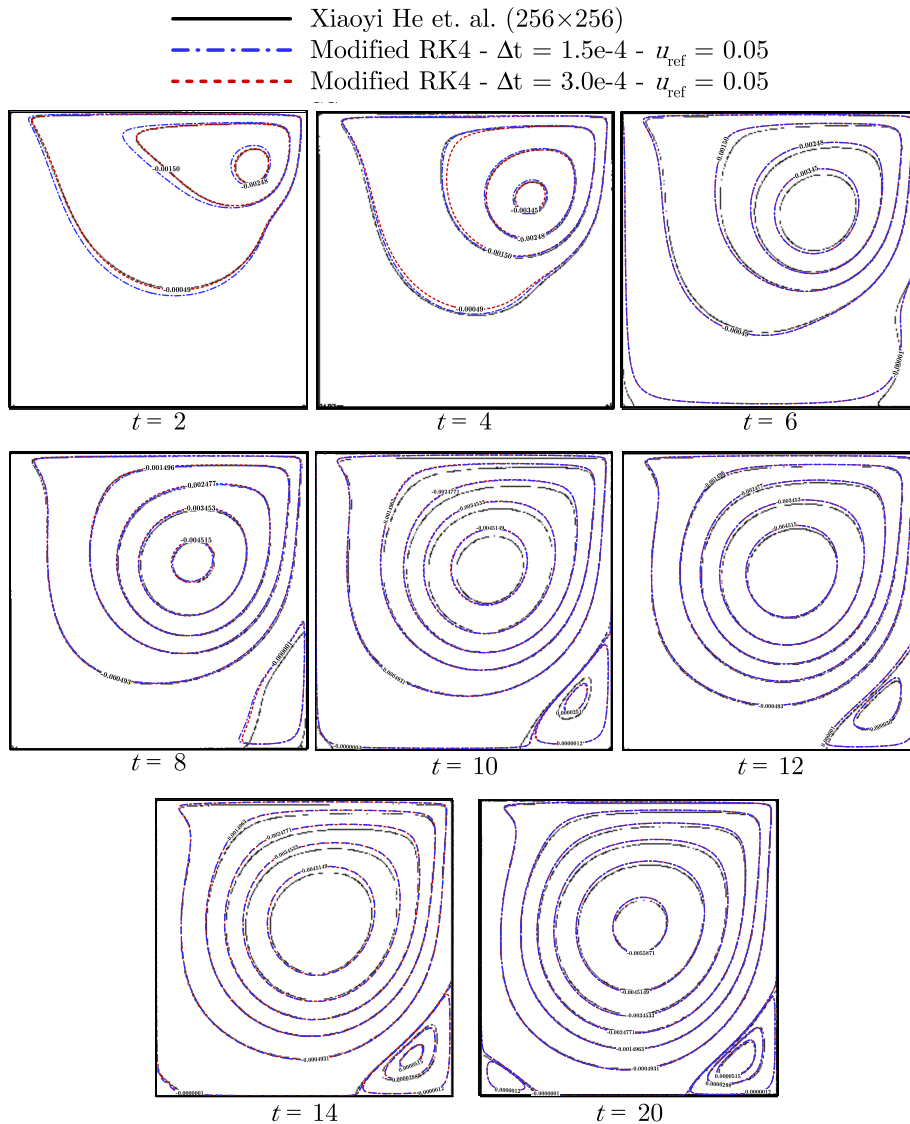


FIG. 5. Comparison of transient solution shown by the streamlines for the 2D cavity flow with  $Re = 1000$ .

obtained straightforwardly by shifting the corresponding indices by  $-1$ .

## V. BOUNDARY CONDITIONS IMPLEMENTATION

The implementation of the boundary conditions for the WENOLBM in the three-dimensional generalized coordinates consists of two parts; at first, the macroscopic properties are implemented at the desired boundary similar to those implemented for the incompressible Navier–Stokes solvers, and then, the microscopic properties should be calculated at that boundary. Note that the distribution function  $f_x$  cannot be calculated directly at each boundary, and therefore, suitable treatments<sup>16,17,21,22</sup> are needed to be implemented there.

At the solid wall and inlet boundaries, where the velocity components of the flow are known, the zero value for the first derivative of

the pressure normal to the boundary surface is used,  $\partial p / \partial x^i \simeq 0$ , and by applying the fourth-order one-sided finite-difference relation, the pressure is determined at the boundary as follows:

$$\begin{aligned} p_1 &= \frac{1}{125}(240p_2 - 180p_3 + 80p_4 - 15p_5), \\ p_{I_{\max}} &= \frac{1}{125}(240p_{I_{\max}-1} - 180p_{I_{\max}-2} + 80p_{I_{\max}-3} - 15p_{I_{\max}-4}). \end{aligned} \quad (20)$$

At an outlet boundary where the pressure field is known, the same procedure can be used to calculate the velocity components at the boundary points.

Now, the way to calculate the microscopic properties at each boundary is described. The distribution function at the boundary is determined as

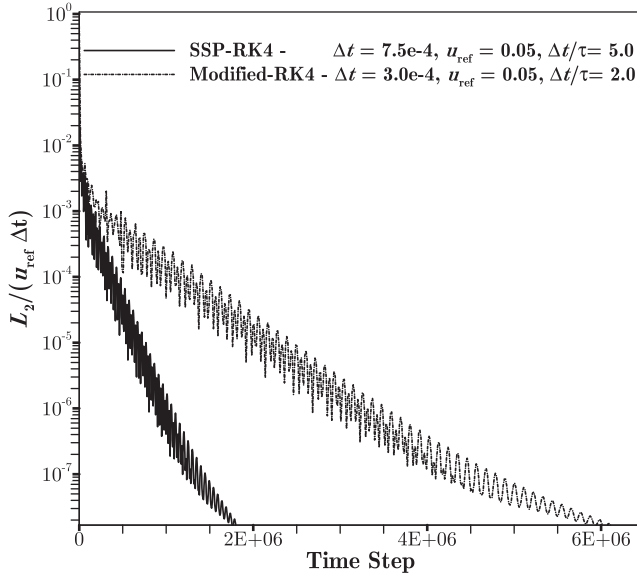


FIG. 6. Comparison of convergence rate of steady-state solution using the SSP-RK4 and modified-RK4 schemes for the 2D cavity flow with  $Re = 1000$ .

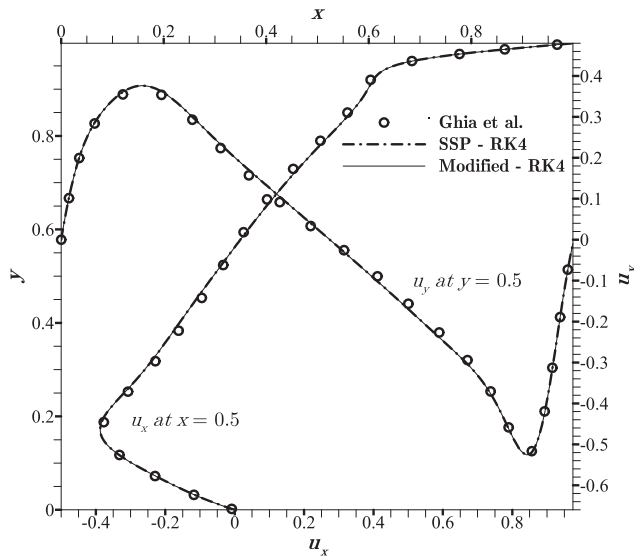


FIG. 7. Comparison of centerline velocity profiles at steady-state condition for the 2D cavity flow with  $Re = 1000$ .

$$f_{\alpha} = f_{\alpha}^{\text{eq}} + f_{\alpha}^{\text{non-eq}}, \quad (21)$$

where the equilibrium distribution function  $f_{\alpha}^{\text{eq}}$  with the known macroscopic properties at the boundary can be calculated using Eq. (2). To calculate the non-equilibrium distribution function  $f_{\alpha}^{\text{non-eq}}$ , a zero value for the second derivative of  $f_{\alpha}^{\text{non-eq}}$  normal to the boundary  $\partial^2 f_{\alpha}^{\text{non-eq}} / (\partial x^i)^2 = 0$  is applied and by using the fourth-order one-sided finite-difference relation,  $f_{\alpha}^{\text{non-eq}}$  can be calculated at the boundary as follows:

$$\begin{aligned} f_{\alpha,1}^{\text{non-eq}} &= \frac{1}{180} \left( 616f_{\alpha,2}^{\text{non-eq}} - 856f_{\alpha,3}^{\text{non-eq}} + 624f_{\alpha,4}^{\text{non-eq}} \right. \\ &\quad \left. - 244f_{\alpha,5}^{\text{non-eq}} + 40f_{\alpha,6}^{\text{non-eq}} \right), \\ f_{\alpha,I_{\text{max}}}^{\text{non-eq}} &= \frac{1}{180} \left( 616f_{\alpha,I_{\text{max}}-1}^{\text{non-eq}} - 856f_{\alpha,I_{\text{max}}-2}^{\text{non-eq}} + 624f_{\alpha,I_{\text{max}}-3}^{\text{non-eq}} \right. \\ &\quad \left. - 244f_{\alpha,I_{\text{max}}-4}^{\text{non-eq}} + 40f_{\alpha,I_{\text{max}}-5}^{\text{non-eq}} \right). \end{aligned} \quad (22)$$

## VI. NUMERICAL RESULTS

For validating and examining the accuracy and robustness of the present high-order WENOLBM solver, different test cases are simulated here. These test cases are as follows: 2D cavity flow, 3D Beltrami flow, 3D cavity flow, 3D flow in the curved ducts with the rectangular cross section, and the flow past the sphere with different flow conditions. In addition, a sensitivity analysis is performed to assess the effects of different numerical parameters on the accuracy and performance of the solution method applied.

### A. 2D cavity flow

A proper benchmark problem for assessing the accuracy and performance of the numerical schemes in the incompressible flows is the lid-driven cavity problem, and due to the simplicity of its geometry and boundary conditions, different studies have been performed in the literature on this problem in both two and three dimensions. Here, the transient incompressible flow inside the two-dimensional cavity at  $Re = u_{\text{ref}}L/\nu = 1000$  with the reference length  $L = 1.0$  is computed by applying the WENOLBM using the D2Q9 discrete velocity model and the accuracy and performance of the modified RK4 and SSP-RK4 time integration schemes implemented are compared with each other. The simulation is performed by the WENOLBM unsteadily until the steady-state solution is achieved. The present solution for this test case obtained by applying the WENOLBM using the two mentioned time integration schemes is compared with that reported by He *et al.*,<sup>64</sup> in which their results are based on the solution of the standard LBM using the uniform grid  $256 \times 256$ . Figure 3 compares the time history of the early transient solution of the  $u$ -velocity variable at the center of the two-dimensional cavity problem. The present results in this figure are obtained based on the modified RK4 scheme considering different values of the reference velocity  $u_{\text{ref}}$ . The study shows that the early transient solution is sensitive to the value of the reference velocity. Note that the value of the reference velocity  $u_{\text{ref}}$  controls the level of the compressibility in the flow domain and thus different values of  $u_{\text{ref}}$  mean different compressibility effects. It is observed that the present solution obtained by applying the WENOLBM with  $u_{\text{ref}} = 0.05$  is reasonably matched with that of He *et al.* The study indicates that the solution of the WENOLBM obtained by the stretched grid  $128 \times 128$  is more accurate than that of the uniform grid  $256 \times 256$ . Figure 4 compares the time history of the early transient solution of the  $u$ -velocity variable at the center of the two-dimensional cavity problem for the two mentioned time integration schemes. The study indicates that the WENOLBM using the SSP-RK4 scheme requires a much smaller time step size for an accurate transient solution and the corresponding computation time is about 30 times higher than that of the modified RK4 scheme (see Table II).

Figure 5 gives the time history of the present solution obtained by the WENOLBM shown by the streamlines for the two-dimensional

**TABLE III.** Order of accuracy of solution based on  $L_2$  norm of velocity profile using the modified RK4 time integration with  $\Delta t/\tau = 2.0$  for the 3D Beltrami flow with  $Re = 25$ .

|                    | Grid                     | $\Delta x$ | $\text{Log}(\Delta x)$ | $L_2$                 | $\text{Log}(L_2)$ |
|--------------------|--------------------------|------------|------------------------|-----------------------|-------------------|
| $u_{ref} = 0.1$    | $9 \times 9 \times 9$    | 0.125      | -0.903 09              | $3.65 \times 10^{-6}$ | -5.4371           |
|                    | $17 \times 17 \times 17$ | 0.062 5    | -1.204 12              | $5.27 \times 10^{-7}$ | -6.2785           |
|                    | $33 \times 33 \times 33$ | 0.031 25   | -1.505 15              | $7.06 \times 10^{-8}$ | -7.1512           |
|                    | $65 \times 65 \times 65$ | 0.015 625  | -1.806 18              | $9.15 \times 10^{-9}$ | -8.0385           |
|                    | Order of accuracy        |            |                        |                       | 2.9232            |
|                    | Grid                     | $\Delta x$ | $\text{Log}(\Delta x)$ | $L_2$                 | $\text{Log}(L_2)$ |
| $u_{ref} = 0.01$   | $9 \times 9 \times 9$    | 0.125      | -0.903 09              | $3.24 \times 10^{-5}$ | -4.4900           |
|                    | $17 \times 17 \times 17$ | 0.062 5    | -1.204 12              | $5.45 \times 10^{-7}$ | -6.2636           |
|                    | $33 \times 33 \times 33$ | 0.031 25   | -1.505 15              | $2.61 \times 10^{-8}$ | -7.5831           |
|                    | $65 \times 65 \times 65$ | 0.015 625  | -1.806 18              | $4.65 \times 10^{-9}$ | -8.3323           |
|                    | Order of accuracy        |            |                        |                       | 3.4360            |
|                    | Grid                     | $\Delta x$ | $\text{Log}(\Delta x)$ | $L_2$                 | $\text{Log}(L_2)$ |
| $u_{ref} = 0.0050$ | $9 \times 9 \times 9$    | 0.125      | -0.903 09              | $8.67 \times 10^{-5}$ | -4.0617           |
|                    | $17 \times 17 \times 17$ | 0.062 5    | -1.204 12              | $1.50 \times 10^{-6}$ | -5.8235           |
|                    | $33 \times 33 \times 33$ | 0.031 25   | -1.505 15              | $1.95 \times 10^{-8}$ | -7.7097           |
|                    | $65 \times 65 \times 65$ | 0.015 625  | -1.806 18              | $1.97 \times 10^{-9}$ | -8.7055           |
|                    | Order of accuracy        |            |                        |                       | 4.7869            |

**TABLE IV.** Order of accuracy of solution based on  $L_2$  norm of velocity profile using the SSP RK4 time integration for the 3D Beltrami flow with  $Re = 25$ .

|                                           | Grid                     | $\Delta x$ | $\text{Log}(\Delta x)$ | $L_2$                 | $\text{Log}(L_2)$ |
|-------------------------------------------|--------------------------|------------|------------------------|-----------------------|-------------------|
| $u_{ref} = 0.1$<br>$\Delta t/\tau = 5.0$  | $9 \times 9 \times 9$    | 0.125      | -0.903 09              | $1.59 \times 10^{-5}$ | -4.7992           |
|                                           | $17 \times 17 \times 17$ | 0.062 5    | -1.204 12              | $2.97 \times 10^{-6}$ | -5.5279           |
|                                           | $33 \times 33 \times 33$ | 0.031 25   | -1.505 15              | <b>Diverged</b>       | <b>Diverged</b>   |
|                                           | $65 \times 65 \times 65$ | 0.015 625  | -1.806 18              | <b>Diverged</b>       | <b>Diverged</b>   |
|                                           | Order of accuracy        |            |                        |                       | 2.4206            |
|                                           | Grid                     | $\Delta x$ | $\text{Log}(\Delta x)$ | $L_2$                 | $\text{Log}(L_2)$ |
| $u_{ref} = 0.01$<br>$\Delta t/\tau = 5.0$ | $9 \times 9 \times 9$    | 0.125      | -0.903 09              | $5.01 \times 10^{-5}$ | -4.3003           |
|                                           | $17 \times 17 \times 17$ | 0.062 5    | -1.204 12              | $5.69 \times 10^{-6}$ | -5.2446           |
|                                           | $33 \times 33 \times 33$ | 0.031 25   | -1.505 15              | $7.92 \times 10^{-7}$ | -6.1011           |
|                                           | $65 \times 65 \times 65$ | 0.015 625  | -1.806 18              | $1.02 \times 10^{-7}$ | -6.9933           |
|                                           | Order of accuracy        |            |                        |                       | 2.9044            |
|                                           | Grid                     | $\Delta x$ | $\text{Log}(\Delta x)$ | $L_2$                 | $\text{Log}(L_2)$ |
| $u_{ref} = 0.01$<br>$\Delta t/\tau = 2.0$ | $9 \times 9 \times 9$    | 0.125      | -0.903 09              | $1.81 \times 10^{-4}$ | -3.7419           |
|                                           | $17 \times 17 \times 17$ | 0.062 5    | -1.204 12              | $2.78 \times 10^{-5}$ | -4.5565           |
|                                           | $33 \times 33 \times 33$ | 0.031 25   | -1.505 15              | $3.52 \times 10^{-7}$ | -6.4539           |
|                                           | $65 \times 65 \times 65$ | 0.015 625  | -1.806 18              | $4.58 \times 10^{-8}$ | -7.3387           |
|                                           | Order of accuracy        |            |                        |                       | 4.6211            |



TABLE IV. (Continued.)

|                                                   | Grid                     | $\Delta x$ | $\log(\Delta x)$ | $L_2$                 | $\log(L_2)$ |
|---------------------------------------------------|--------------------------|------------|------------------|-----------------------|-------------|
| $u_{\text{ref}} = 0.005$<br>$\Delta t/\tau = 5.0$ | $9 \times 9 \times 9$    | 0.125      | -0.903 09        | $1.23 \times 10^{-4}$ | -3.9089     |
|                                                   | $17 \times 17 \times 17$ | 0.062 5    | -1.204 12        | $8.19 \times 10^{-6}$ | -5.0865     |
|                                                   | $33 \times 33 \times 33$ | 0.031 25   | -1.505 15        | $1.13 \times 10^{-6}$ | -5.9454     |
|                                                   | $65 \times 65 \times 65$ | 0.015 625  | -1.806 18        | $1.45 \times 10^{-7}$ | -6.8396     |
| Order of accuracy                                 |                          |            |                  |                       | 2.9119      |

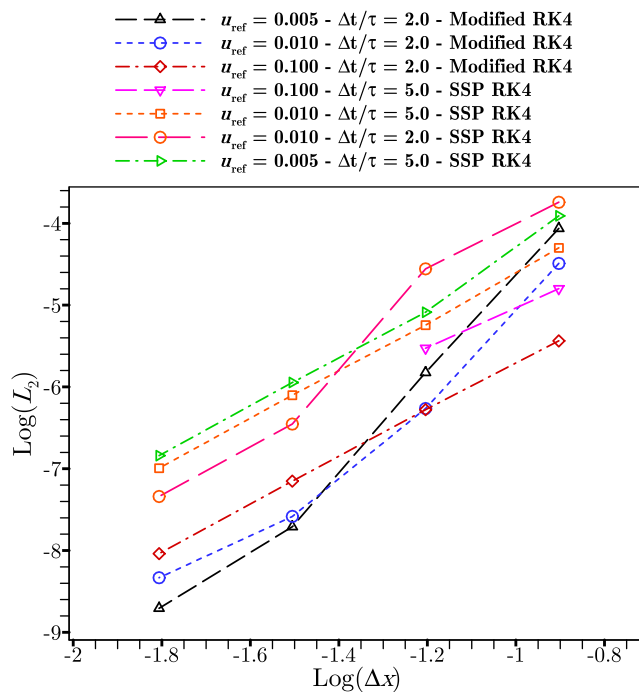
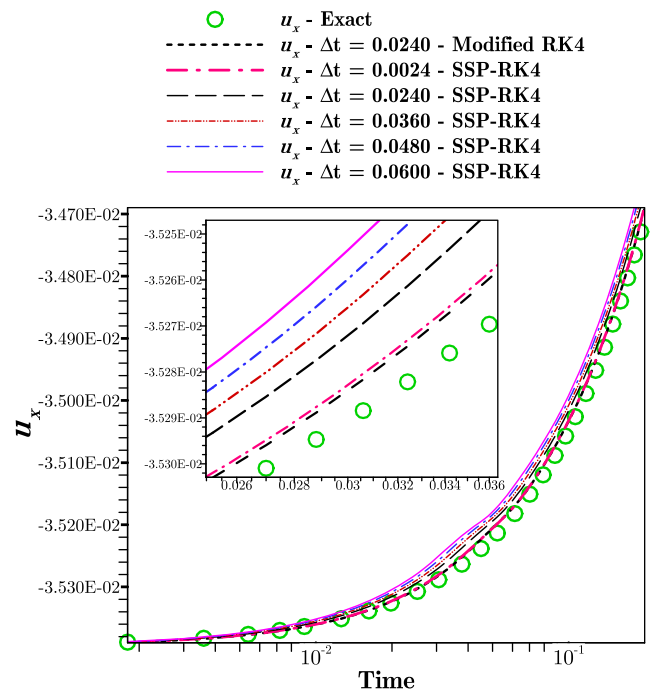


FIG. 8. Comparison of order of accuracy of solution using the SSP-RK4 and modified RK4 schemes for the Beltrami flow.

cavity flow and it is indicated that the transient solution with applying the modified RK4 scheme using the two time step sizes considered,  $\Delta t = 1.5 \times 10^{-4}$  and  $3.0 \times 10^{-4}$  are nearly the same and they are, in general, in good agreement with those of He *et al.*<sup>64</sup> Small differences between the results of these two solution setups observed at the moderate times (i.e.,  $t = 10$ ). The study shows that the results of these two solution setups are better matched with each other by further advancing the time to the steady-state condition and their steady-state results are almost identical.

The convergence history of the steady-state solution for this test case by applying the WENOLBM using the SSP-RK4 and modified RK4 schemes with their maximum allowable time step sizes are shown in Fig. 6. Table II also compares the computation time for these time integration schemes implemented in the WENOLBM. The performance of the WENOLBM with the modified RK4 is much better than the SSP-RK4 scheme for an accurate transient solution; however, the SSP-RK4 scheme provides a wider bound of stability based on the ratio of

FIG. 9. Time history of solution of the  $u_x$ -velocity variable at  $(x, y, z) = (1, 1, 1)$  using the SSP-RK4 and modified RK4 schemes for the Beltrami flow.

the time step size to the relaxation time, i.e.,  $\Delta t/\tau$ , to obtain the steady-state solution. Although the computational cost of the WENOLBM with the SSP-RK4 scheme per each iteration is much higher than with the modified RK4 scheme, the total computation time of the WENOLBM with the SSP-RK4 scheme is about 20% lower than the modified RK4 scheme for the steady-state solution. Figure 7 presents the steady-state solution of the centerline velocity profiles for the two-dimensional cavity flow obtained by applying the WENOLBM using the modified RK4 and SSP-RK4 schemes and the corresponding results are compared with those reported by Ghia *et al.*<sup>65</sup> The study indicates that the steady-state results obtained by implementing these time integration schemes to the WENOLBM are the same and they are in good agreement with those presented in Ref. 65.

### B. 3D Beltrami flow

To examine the accuracy of the solution of the presented three-dimensional high-order WENOLBM, the three-dimensional Beltrami

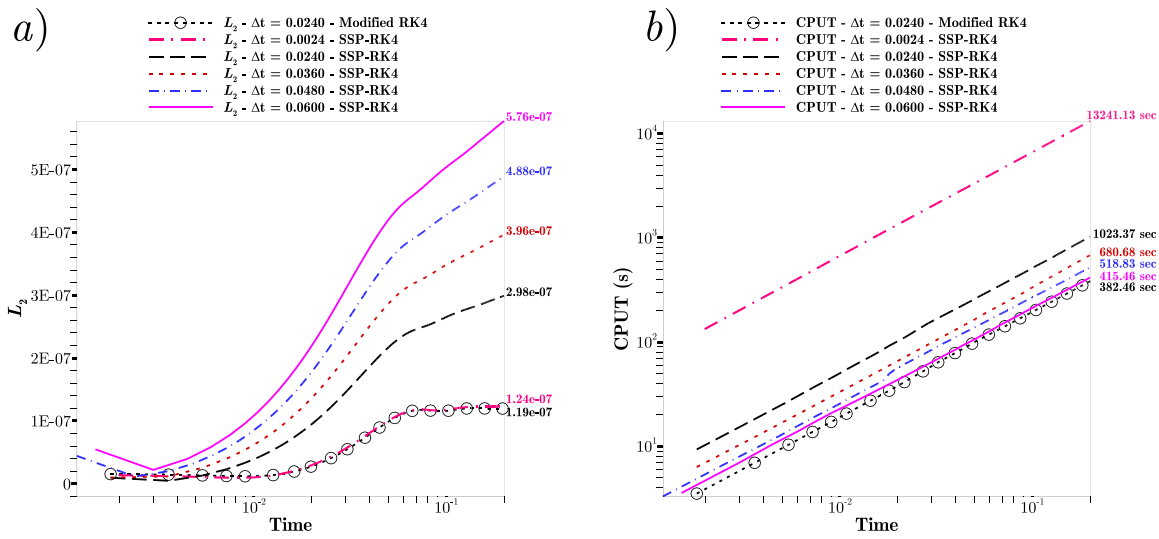


FIG. 10. Time history of (a) solution error based on  $L_2$  norm and (b) total computational time using the SSP-RK4 and modified RK4 schemes for the Beltrami flow.

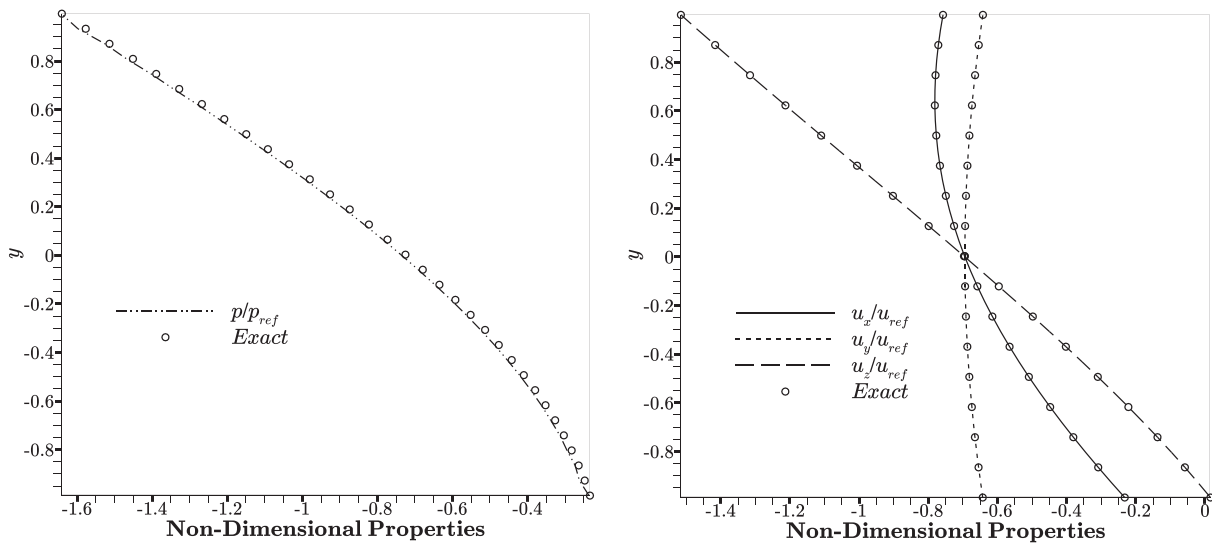


FIG. 11. Comparison of velocity (right) and pressure (left) profiles at midplane  $(0, y, 0)$  for the Beltrami flow with  $Re = 25$  at  $t = 0.1$ .

flow<sup>66</sup> is simulated here in which it has the following analytical solution to be used for validating the present results:

$$\begin{aligned} u(x, y, z, t) &= -u_{ref} a \left( e^{\frac{ax}{L}} \sin\left(\frac{ay}{L} + \frac{dz}{L}\right) + e^{\frac{az}{L}} \cos\left(\frac{ax}{L} + \frac{dy}{L}\right) \right) e^{-\frac{u_{ref}^2 t}{L^2}}, \\ v(x, y, z, t) &= -u_{ref} a \left( e^{\frac{ay}{L}} \sin\left(\frac{az}{L} + \frac{dx}{L}\right) + e^{\frac{ax}{L}} \cos\left(\frac{ay}{L} + \frac{dz}{L}\right) \right) e^{-\frac{u_{ref}^2 t}{L^2}}, \\ w(x, y, z, t) &= -u_{ref} a \left( e^{\frac{az}{L}} \sin\left(\frac{ax}{L} + \frac{dy}{L}\right) + e^{\frac{ay}{L}} \cos\left(\frac{az}{L} + \frac{dx}{L}\right) \right) e^{-\frac{u_{ref}^2 t}{L^2}}, \end{aligned} \quad (23)$$

$$\begin{aligned} p(x, y, z, t) &= -\rho \frac{u_{ref}^2 a^2}{2} \left( e^{\frac{2ax}{L}} + e^{\frac{2ay}{L}} + e^{\frac{2az}{L}} + 2 \sin\left(\frac{ax}{L} + \frac{dy}{L}\right) \right. \\ &\quad \times \cos\left(\frac{az}{L} + \frac{dx}{L}\right) e^{a\left(\frac{x}{L} + \frac{y}{L}\right)} \\ &\quad + 2 \sin\left(\frac{ay}{L} + \frac{dz}{L}\right) \cos\left(\frac{ax}{L} + \frac{dy}{L}\right) e^{a\left(\frac{y}{L} + \frac{z}{L}\right)} \\ &\quad \left. + 2 \sin\left(\frac{az}{L} + \frac{dx}{L}\right) \cos\left(\frac{ay}{L} + \frac{dz}{L}\right) e^{a\left(\frac{z}{L} + \frac{x}{L}\right)} \right) \\ &\quad \times e^{-\frac{2u_{ref}^2 t}{L^2}}, \end{aligned}$$

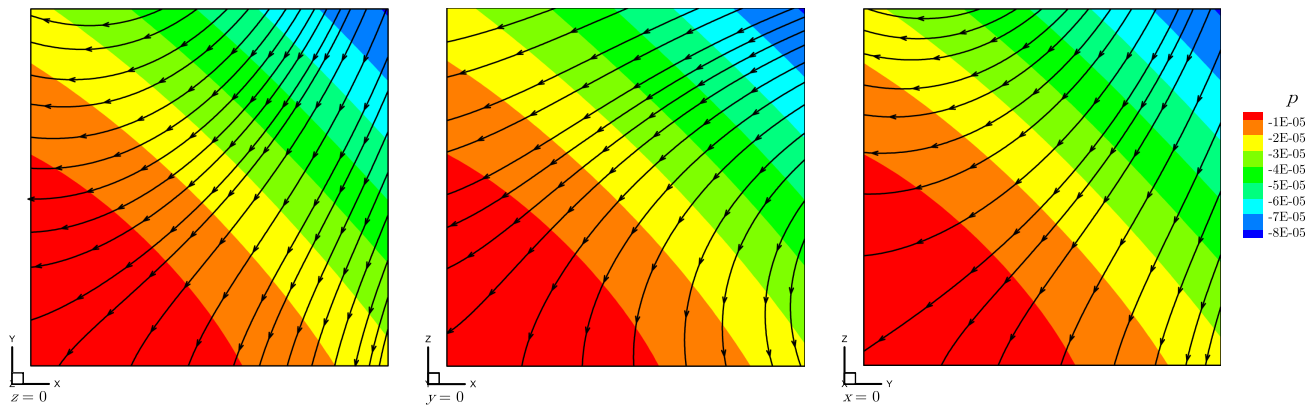


FIG. 12. Computed flow field shown by pressure contours and streamlines at midplanes for the Beltrami flow with  $Re = 25$  at  $t = 0.1$ .

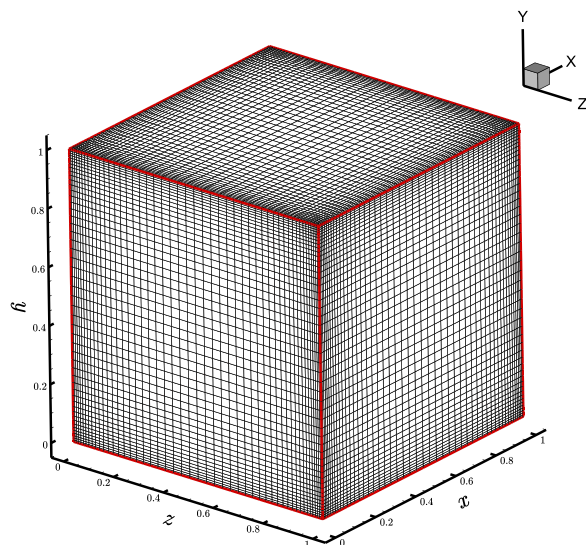


FIG. 13. Clustered grid system ( $51 \times 51 \times 51$ ) generated for the 3D cavity flow.

where  $a$  and  $d$  are the arbitrary coefficients, and here, they are set to  $\pi/4$  and  $\pi/2$ , respectively. This problem is simulated for a cube in the Cartesian coordinates, in which  $-L/2 \leq x, y, z \leq L/2$ , and the center placed at the origin of the coordinate. This problem is initialized using the relations (23) at  $t = 0$ , and the flow parameters at all the boundaries are imposed using the corresponding exact values. The computations are performed for the Reynolds number  $Re = u_{ref}L/\nu = 25$  with the reference length  $L = 2.0$  and considering the constant time step size  $\Delta t = 10^{-3}$  until  $t = 0.1$ . Four uniform grids,  $(9 \times 9 \times 9)$ ,  $(17 \times 17 \times 17)$ ,  $(33 \times 33 \times 33)$ , and  $(65 \times 65 \times 65)$  are used for verifying the order of accuracy of the solution of the WENOLBM considering different values of the reference velocity  $u_{ref}$  by applying the modified RK4 and SSP-RK4 time integration schemes, as given in Tables III and IV, respectively, and also plotted in Fig. 8. Here, the

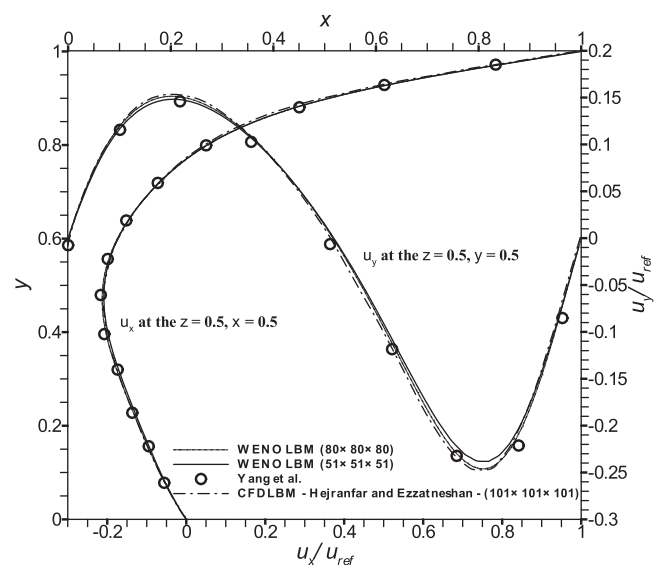


FIG. 14. Comparison of centerline velocity profiles for the 3D cavity flow with  $Re = 100$ .

solution error of the WENOLBM is calculated based on the  $L_2$  norm of the velocity profile compared with the analytical solution of the incompressible flow. The results indicate that the desired order of the accuracy of the solution of the WENOLBM for both the modified RK4 and SSP-RK4 schemes can be achieved by considering a small reference velocity that causes the compressibility effects to be reduced. It is also observed that the level of the solution error of the WENOLBM with the modified RK4 scheme is much smaller than that of the SSP-RK4 scheme.

To further examine the accuracy of the WENOLBM with applying the two time integration schemes for the three-dimensional Beltrami flow, the  $x$ -component of the velocity field at the center of the geometry  $(x, y, z) = (0, 0, 0)$  is extracted in term of the scaled time  $t = tu_{ref}/L$ , as shown in Fig. 9. Here, the solution of the WENOLBM

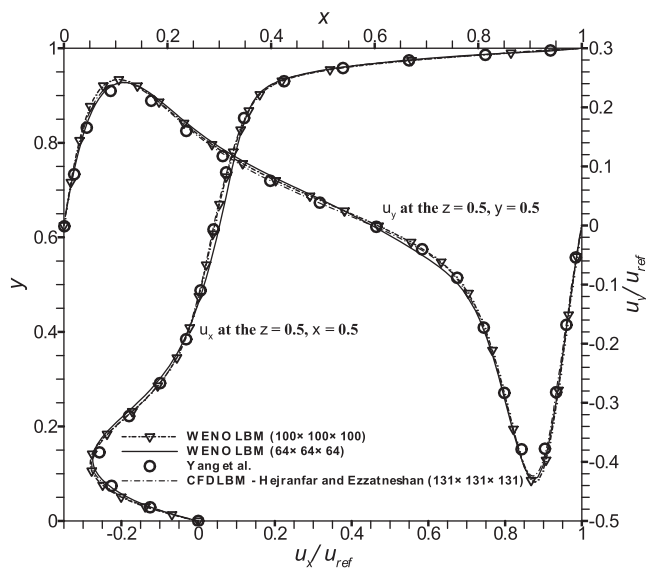


FIG. 15. Comparison of centerline velocity profiles for the 3D cavity flow with  $Re = 1000$ .

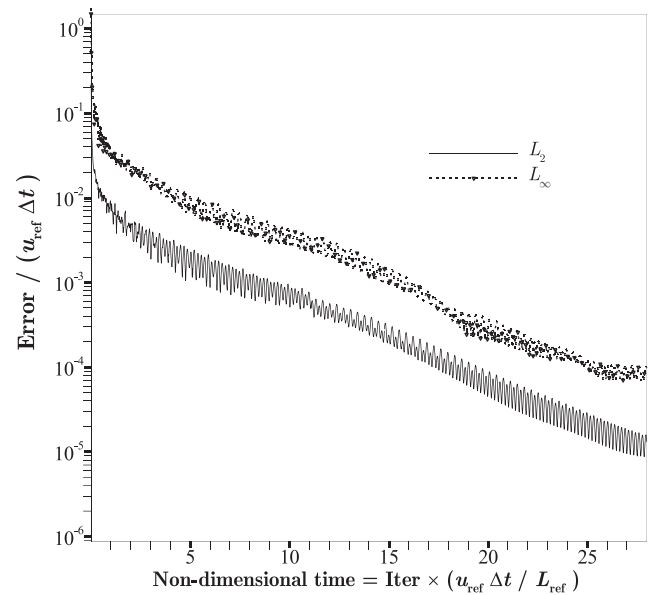


FIG. 18. History of convergence rate of solution for the 3D cavity flow with  $Re = 1000$ .

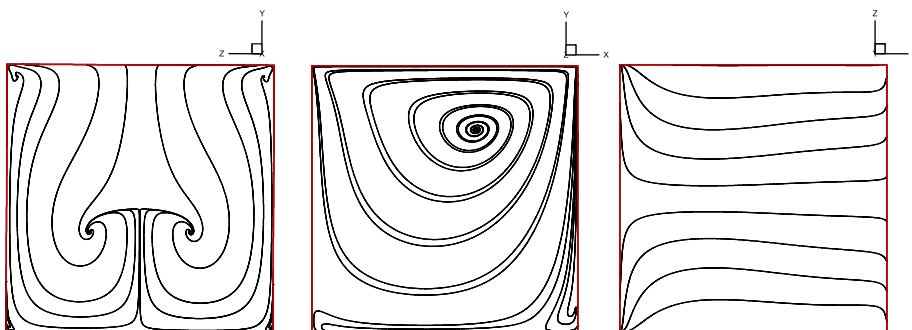


FIG. 16. Computed flow field shown by the streamlines at midplanes  $x = 0.5$  (left),  $z = 0.5$  (middle), and  $y = 0.5$  (right) for the 3D cavity flow with  $Re = 100$ .

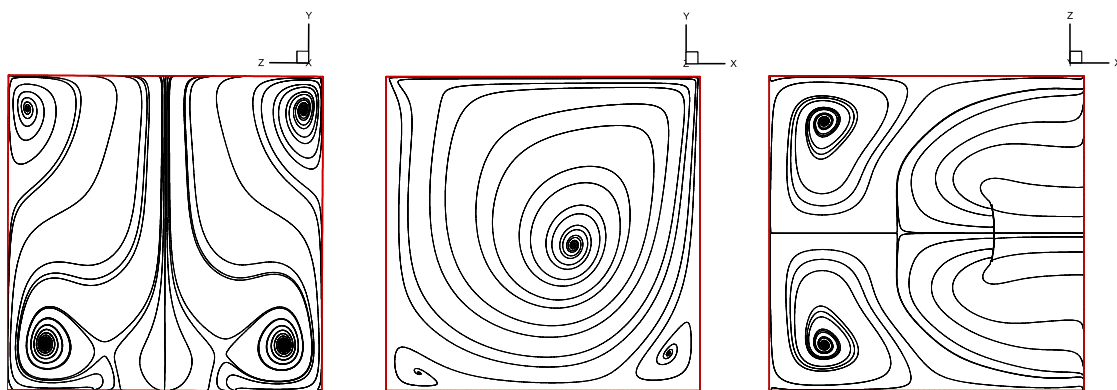


FIG. 17. Computed flow field shown by the streamlines at midplanes  $x = 0.5$  (left),  $z = 0.5$  (middle), and  $y = 0.5$  (right) for the 3D cavity flow with  $Re = 1000$ .

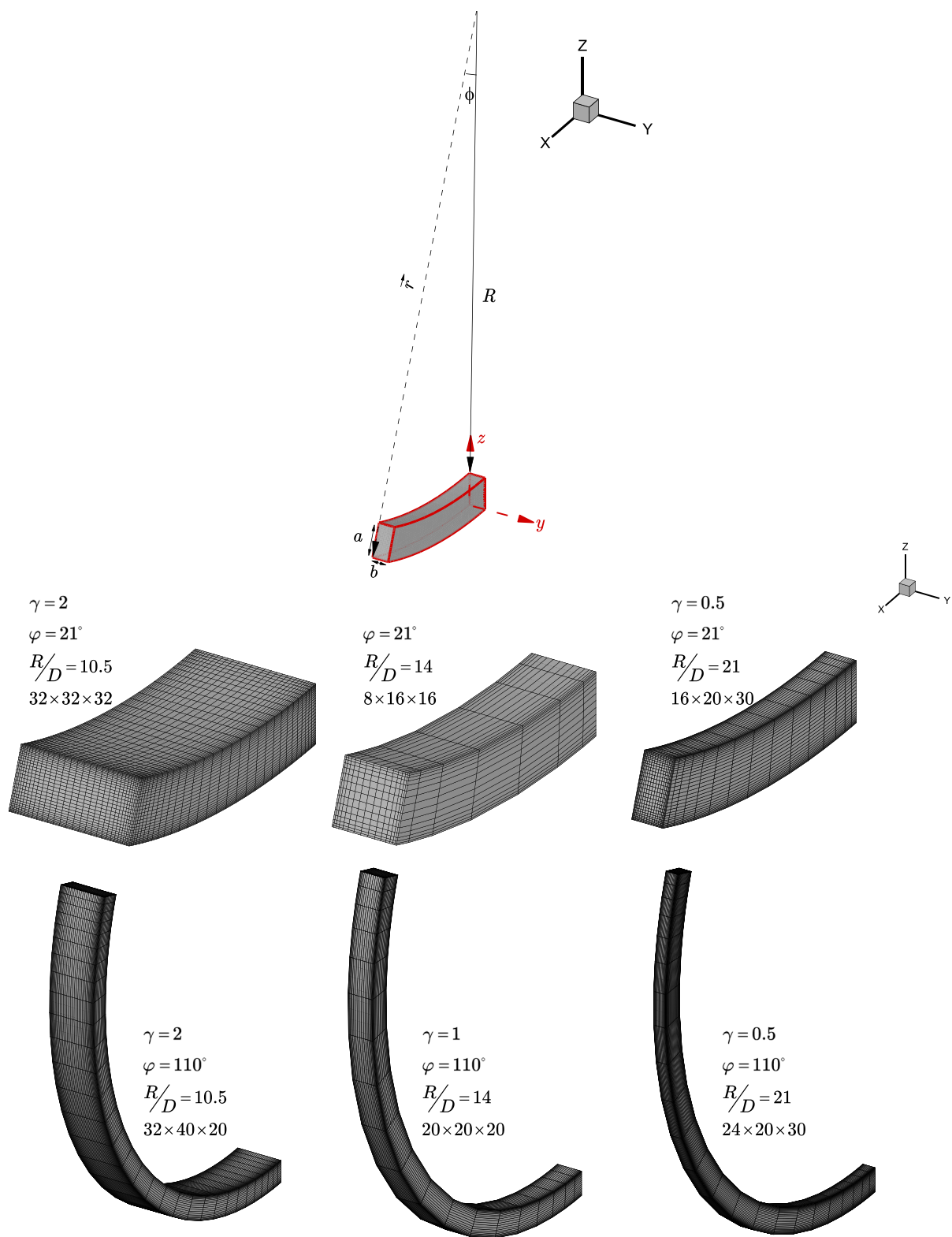


FIG. 19. Geometry detail of 3D curved duct and generated grids for six configurations.

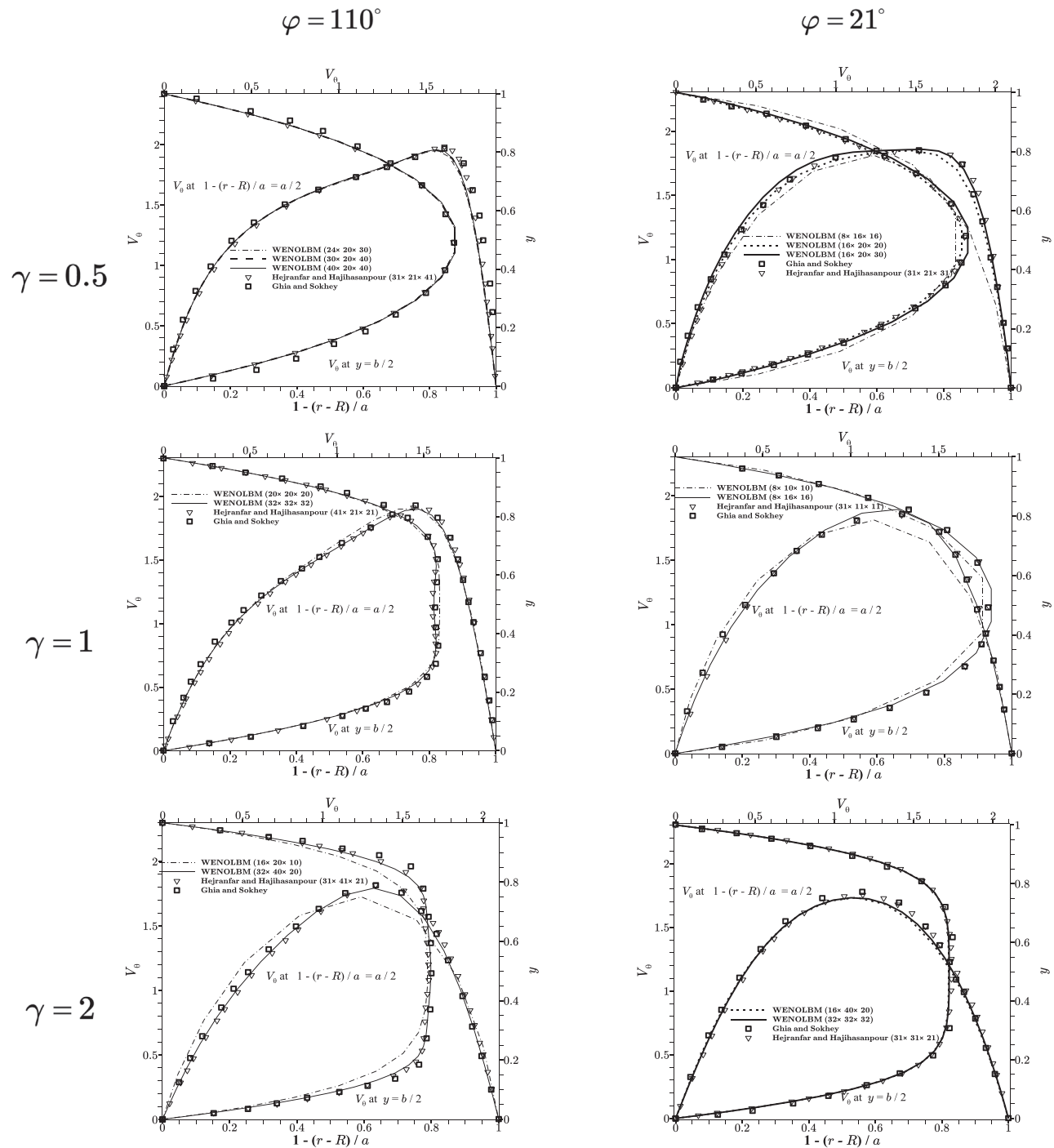
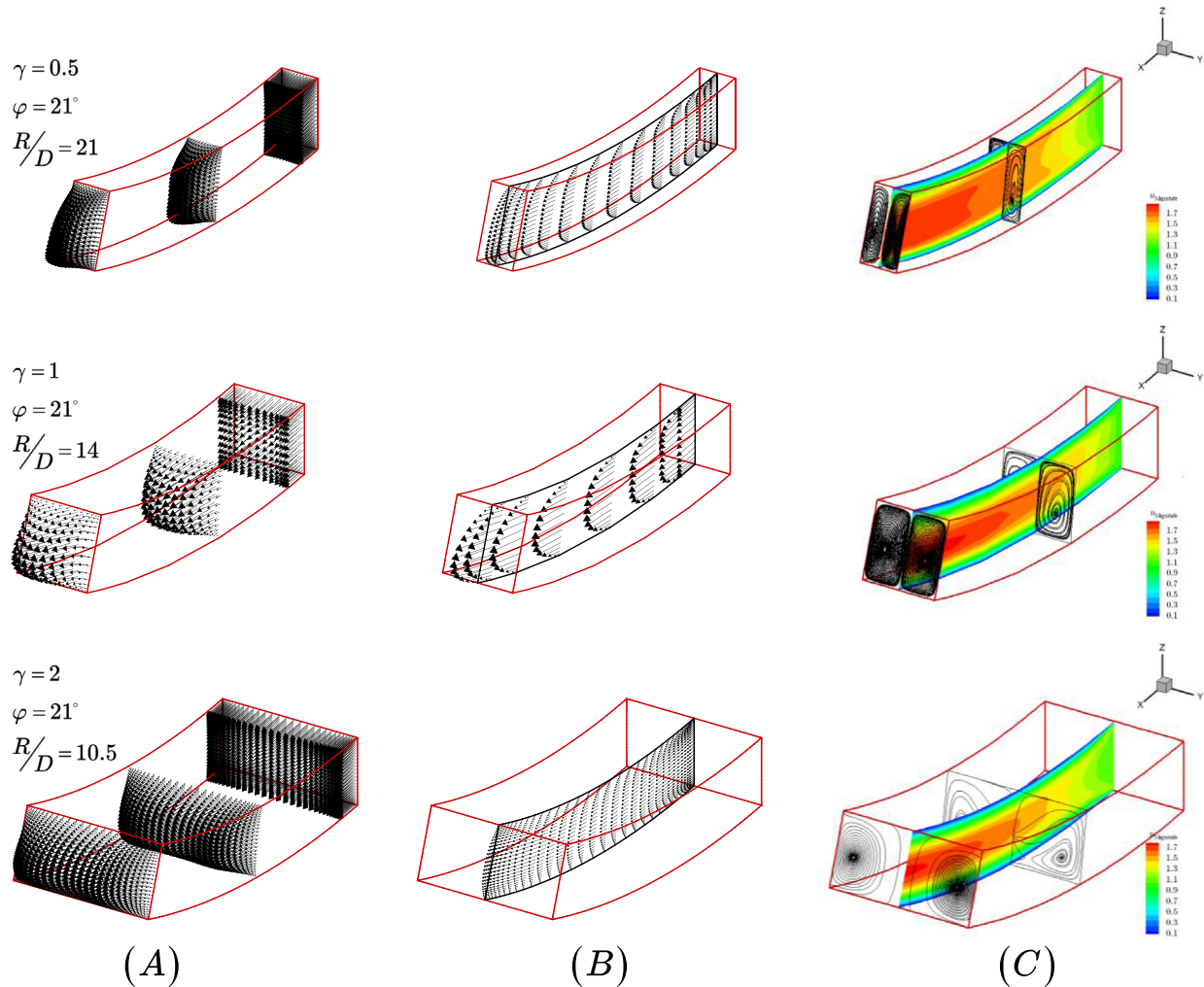


FIG. 20. Comparison of axial velocity profiles at outlet midplanes of the 3D curved duct for six configurations.

with the SSP-RK4 time integration scheme considering different time step sizes  $\Delta t = 0.0024, 0.024, 0.036, 0.048, 0.06$  (equivalent with  $\Delta t/\tau = 0.2, 2.0, 3.0, 4.0, 5.0$ ) is compared with that of the modified RK4 with  $\Delta t = 0.024$  (equivalent with  $\Delta t/\tau = 2.0$ ). Figure 10 also

presents the time history of the corresponding solution error and the required computational time for each condition. The study indicates that the WENOLBM with the modified RK4 scheme provides more accurate solution than that of the SSP-RK4 scheme considering a same





**FIG. 21.** Velocity profiles at three sections along the 3D curved duct (a) and in symmetry plane (b) with sectional flow field and velocity contours in symmetry plane (c) for three amounts of aspect ratio  $\gamma = 0.5, 1, 2$  (rows) and rotation angle  $\phi = 21^\circ$

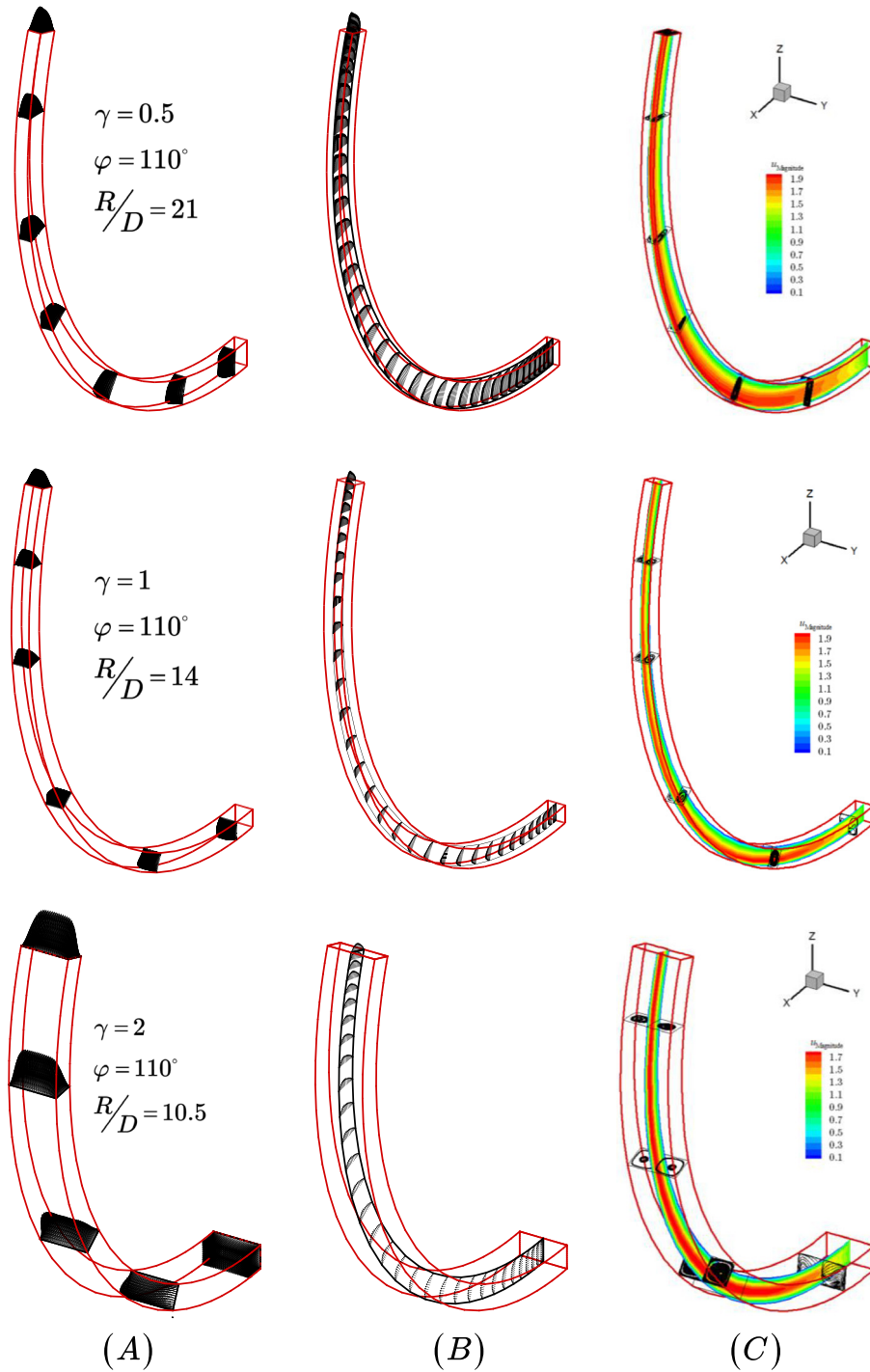
time step size, i.e.,  $\Delta t = 0.024$  ( $\Delta t/\tau = 2.0$ ). To have the same accuracy of the solution, a smaller time step size, i.e.,  $\Delta t = 0.0024$  ( $\Delta t/\tau = 0.2$ ), should be considered for the WENOLBM with the SSP-RK4 scheme, which considerably increases the computational cost of the solution compared to the modified RK4 scheme. Considering a same time step size, i.e.,  $\Delta t = 0.024$  ( $\Delta t/\tau = 2.0$ ), the solution of the WENOLBM with the modified RK4 scheme is about 2.7 times faster than that of the SSP-RK4 scheme and this value is about 34.6 considering a same accuracy of the solution.

Figure 11 gives the non-dimensional velocity and pressure profiles at the midline  $(0, y, 0)$  obtained by applying the WENOLBM with the modified RK4 scheme considering the reference velocity  $u_{\text{ref}} = 0.0050$  compared with the analytical results. Note that these flow profiles at the other midlines are similar, and thus, they are not presented here. It is observed that the present results are in good agreement with the analytical solution. Figure 12 shows the pressure

contours and the streamlines for the midplanes to indicate the behavior of the variation of the flow field in the flow domain. The study indicates that the solution methodology presented based on the high-order WENOLBM provides accurate and reliable results in three dimensions. Note that all the subsequent computations by applying the WENOLBM are based on the modified RK4 scheme.

### C. 3D cavity flow

Here, the lid-driven cavity problem is simulated in three dimensions at the two Reynolds numbers  $Re = u_{\text{ref}}L/\nu = 100, 1000$  for a cube by the edge size of  $L = 1.0$  and the accuracy of the present three-dimensional high-order WENOLBM is assessed through comparing the present solution with the available numerical results. All the boundary surfaces are stationary indicating that the velocity components are zero, except that the  $u$ -component of the boundary surface



**FIG. 22.** Velocity profiles at three sections along the 3D curved duct (a) and in symmetry plane (b) with sectional flow field and velocity contours in symmetry plane (c) for three amounts of aspect ratio  $\gamma = 0.5, 1, 2$  (rows) and rotation angle  $\phi = 110^\circ$

with the coordinates ( $0 \leq \bar{x} \leq 1, \bar{y} = 1, 0 \leq \bar{z} \leq 1$ ) is equal to the reference velocity  $u_{ref} = 0.1$  and the other velocity components are zero on this boundary surface. For resolving the flow gradients near the wall boundaries in the  $x$  direction, the grid points near the walls are appropriately clustered by using the following transformation.<sup>67</sup>

$$\bar{x}_i = \frac{x_i}{L} = \frac{(2\alpha + \beta)[(1 + \beta)/(\beta - 1)]^{(\bar{\xi}_i - \alpha)/(1 - \alpha)} + 2\alpha - \beta}{(2\alpha + 1)\{1 + (\beta + 1)/(\beta - 1)\}^{(\bar{\xi}_i - \alpha)/(1 - \alpha)}},$$

$$0 \leq \bar{\xi}_i = \frac{i - 1}{I_{max} - 1} \leq 1, \quad (24)$$

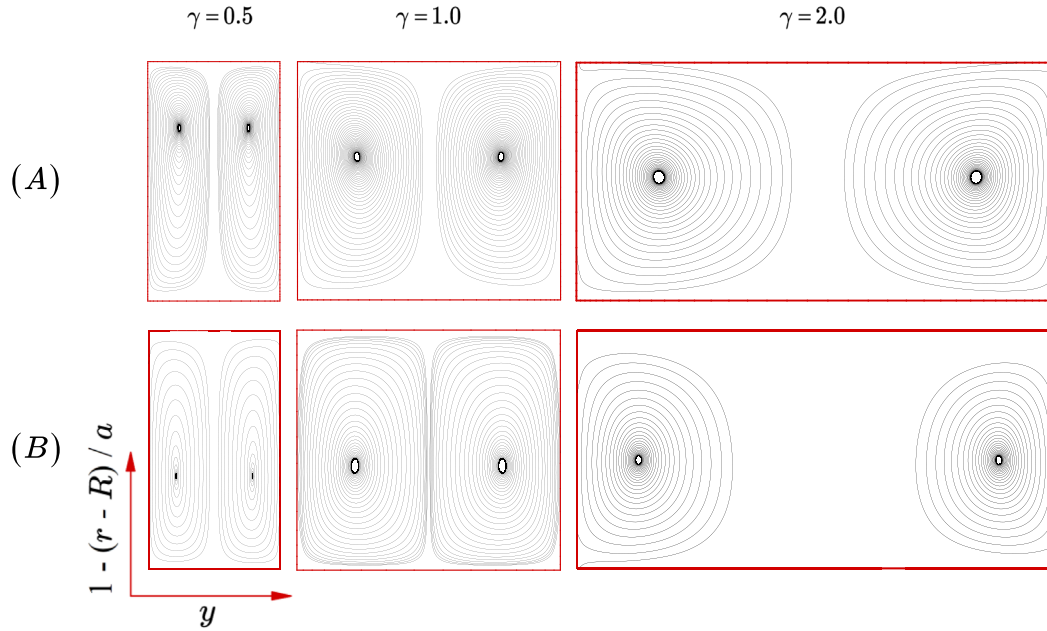


FIG. 23. Velocity field at outlet for rotation angle  $\phi = 21^\circ$  (a) and  $\phi = 110^\circ$  (b) and three values of aspect ratio  $\gamma = 0.5, 1, 2$  (columns).

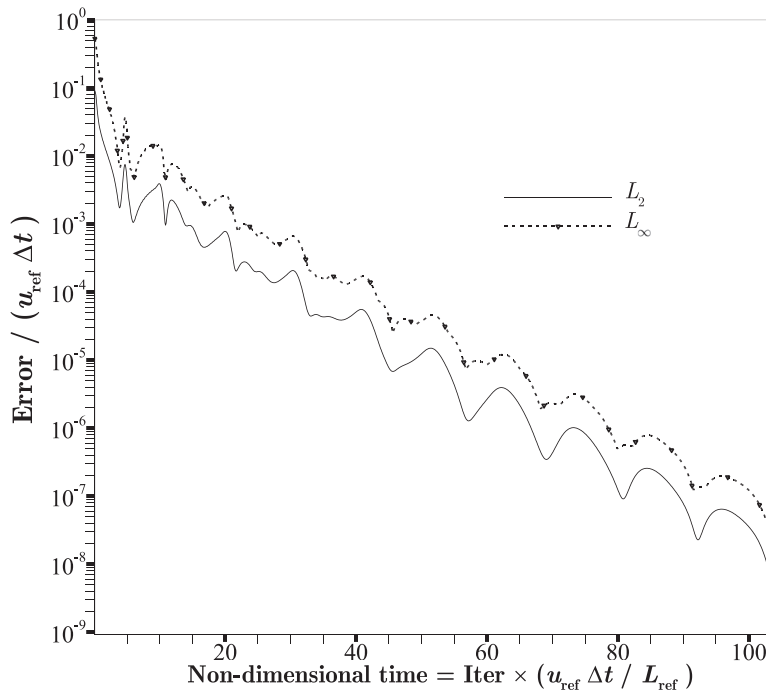


FIG. 24. History of convergence rate of solution for flow in the 3D curved duct—case of  $\phi = 110^\circ$  and  $\gamma = 2$ .

where  $\beta$  is the clustering parameter and the parameter  $\alpha$  specifies where the clustering occurs. The same transformation is used for clustering the grid points near the walls in the  $y$  and  $z$  directions. A grid system with the grid points  $(51 \times 51 \times 51)$  generated by using this transformation considering  $\beta = 1.05$  and  $\alpha = 0.5$  is shown in Fig. 13.

Here, this test case is simulated in two grid systems  $(51 \times 51 \times 51)$  and  $(80 \times 80 \times 80)$  for  $\text{Re} = 100$  and  $(64 \times 64 \times 64)$  and  $(100 \times 100 \times 100)$  for  $\text{Re} = 1000$ . For validating the computed results, the midplanes velocity components at the centerlines  $(x, 0.5, 0.5)$  and  $(0.5, y, 0.5)$  are compared in Figs. 14 and 15 for the  $\text{Re} = 100$  and

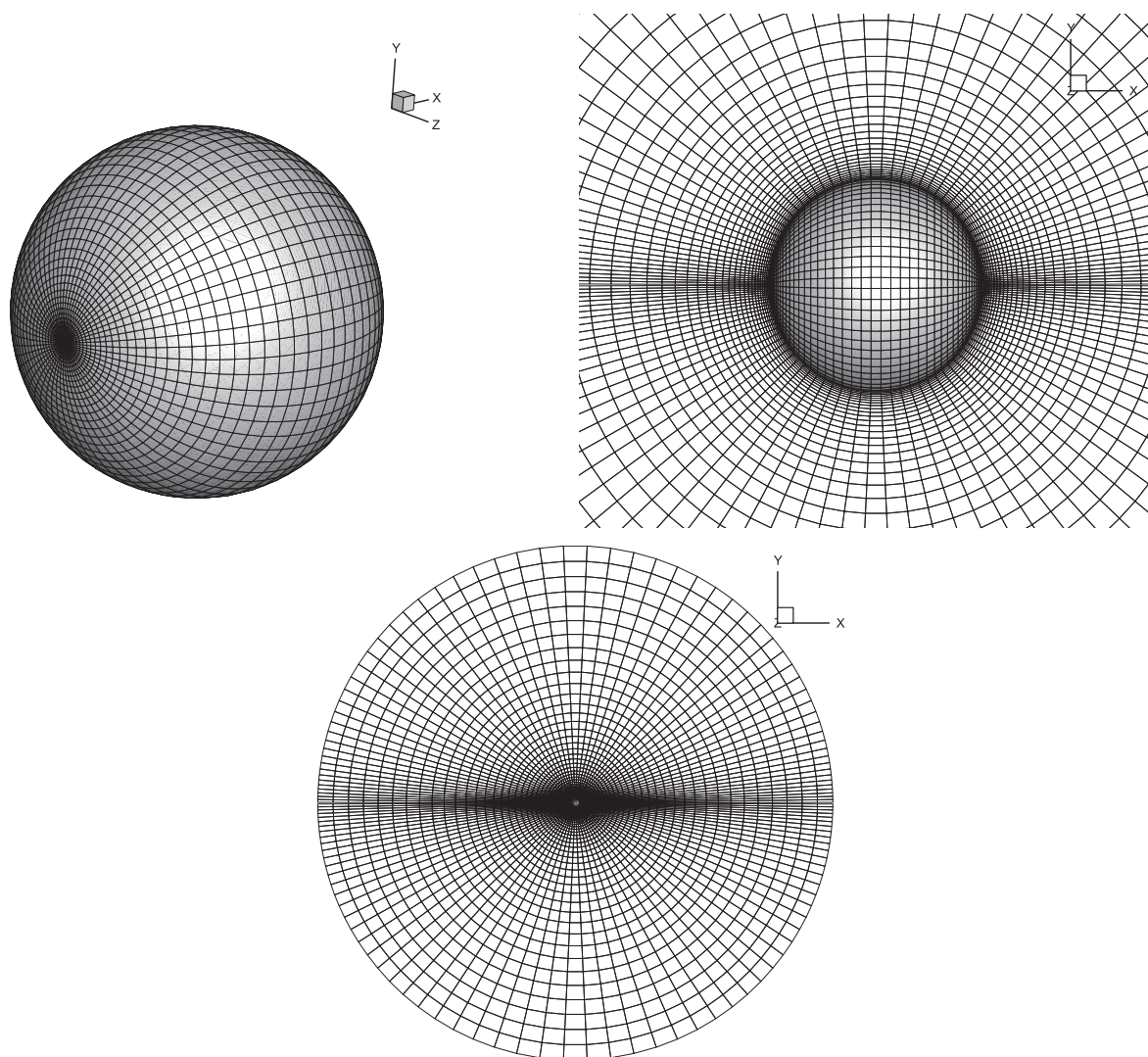


FIG. 25. Generated grid system with  $64 \times 64 \times 64$  grid points for studying flow over the sphere.

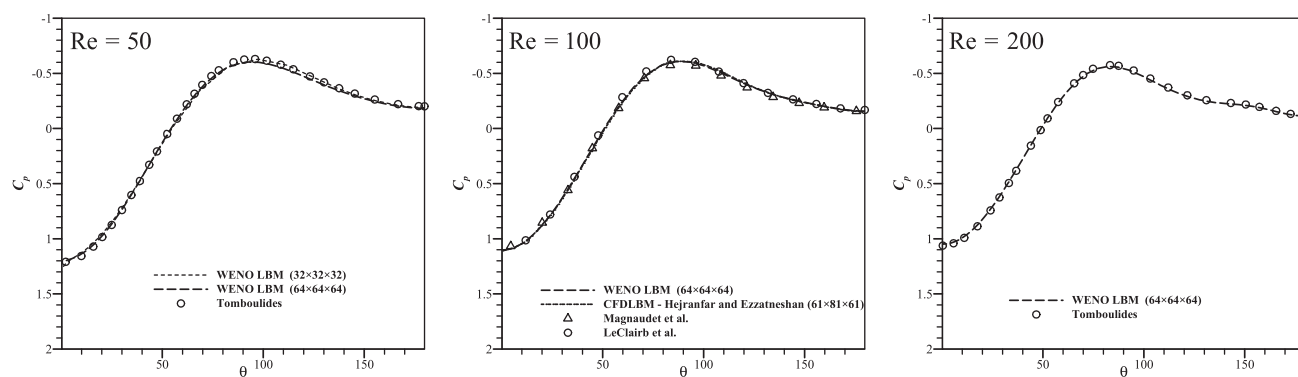


FIG. 26. Comparison (only with numerical references) of surface pressure coefficient distribution for flow over the sphere with  $Re = 50, 100$ , and  $200$  (from left to right).

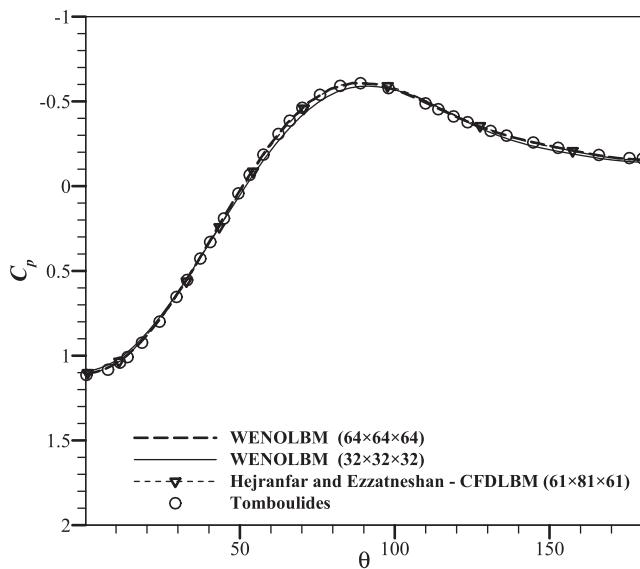


FIG. 27. Comparison of surface pressure coefficient distribution for flow over the sphere with  $Re = 100$ .

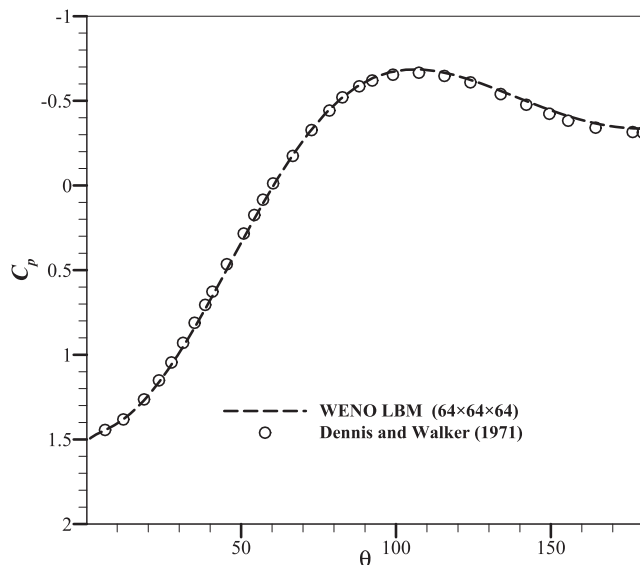


FIG. 28. Comparison of surface pressure coefficient distribution for flow over the sphere with  $Re = 20$ .

1000, respectively, with the results provided by Yang *et al.*<sup>55</sup> and Hejranfar and Ezzatneshan.<sup>68</sup> This comparison shows that the 3D WENOLBM can reasonably resolve the features of the incompressible flows with the coarser grids in comparison with the fourth-order compact finite-difference LBM (CFDLBM).<sup>68</sup>

Figures 16 and 17 show the computed flow field for the 3D cavity problem with  $Re = 100$  and  $Re = 1000$ , respectively. By increasing the Reynolds number, the main circulation in the plane  $z = 0.5$  moves to the center of the geometry and small circulation regions extend in

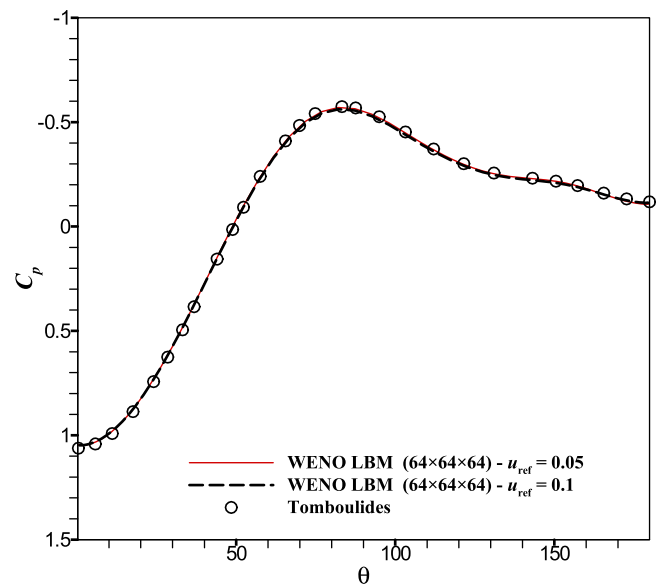


FIG. 29. Comparison of surface pressure coefficient distribution for flow over the sphere with  $Re = 200$  for  $u_{ref} = 0.05$  and  $0.1$ .

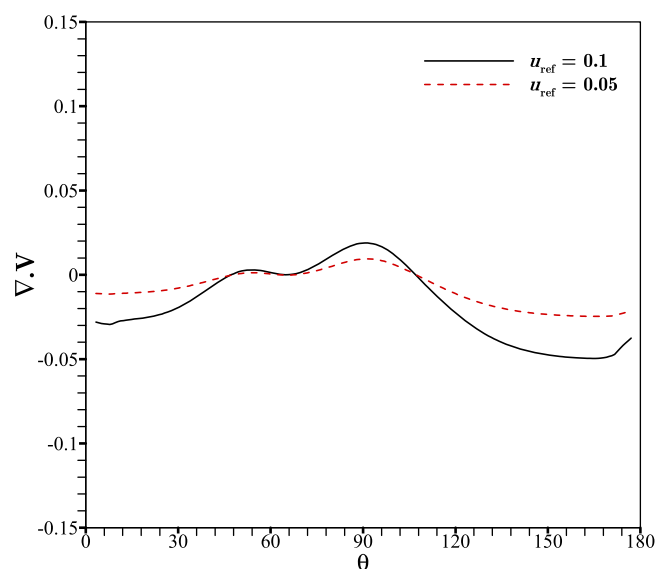


FIG. 30. Comparison of divergence of velocity field on the sphere surface at the steady-state condition with  $Re = 200$  for  $u_{ref} = 0.05$  and  $0.1$ . with  $Re = 200$  for  $u_{ref} = 0.05$  and  $0.1$ .

the bottom corners. It is observed that the 3D flow patterns for  $Re = 1000$  is more complex than that of  $Re = 100$ ; by increasing  $Re$  due to reducing the viscous effects, the two circulation regions near the center of the plane  $x = 0.5$  tend to the bottom corners, and at the same time, the small circulation regions grow symmetrically. The same growths of the circulation regions occur in the midplane of  $y = 0.5$ .

The convergence history of the solution of the 3D WENOLBM solver based on the  $L_2$  and  $L_\infty$  norms of the  $u$ -velocity field for



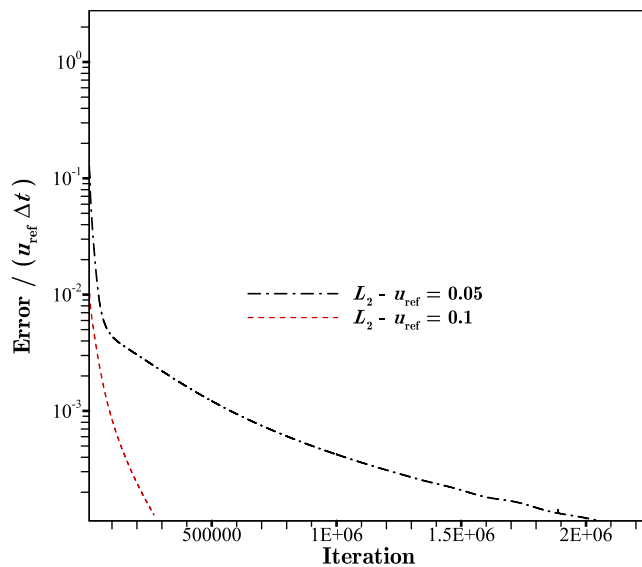


FIG. 31. Convergence history of solution for flow over the sphere with  $Re = 200$  for  $u_{ref} = 0.05$  and  $0.1$ .

$Re = 1000$  is shown in Fig. 18 and it is indicated that the solution properly converges to the steady-state condition. The present results show that the 3D WENOLBM can be considered as an accurate and robust incompressible flow solver with more simplicity in the formulation and implementation compared to the incompressible Navier–Stokes solvers.

### D. 3D flow in curved ducts of rectangular cross-sectional duct

One of the more realistic benchmarks for examining the flow solvers is the flow through a 3D curved duct, and there are available

numerical results for the sake of comparison for this challenging test case. The complexity of the flow features due to the centrifugal forces can be considered as a challenge and a proper grid resolution should be used and the boundary conditions should be implemented in a suitable manner for an accurate solution to be obtained. The study about the centrifugal forces is expressed in term of the Dean number  $K$  that is defined based on the flow Reynolds number and the duct curvature ratio  $R/D$  as  $K = Re/\sqrt{D/R}$ , where  $D$  is the diameter of the duct, and  $R$  is the radius of the duct curvature from the external center of centrifugal forces. For the present rectangular cross-sectional duct, the hydrodynamic diameter  $D_h = 2ab/(a+b)$  defined based on the sectional width  $a$  and length  $b$  (see Fig. 19) is used as the equivalent length characteristics for defining the Reynolds number as  $Re = u_{ref}D_h/\nu$ , where  $u_{ref}$  is the mean flow velocity at the inlet of the duct.

Here, the algebraic grid generation is performed in the cylindrical coordinates  $(r, \phi, y)$ . This test case is simulated for six configurations based on different values for the curvature angle  $\phi$ , the curvature ratio  $R/D$ , and the aspect ratio  $\gamma$ . Figure 19 shows these configurations for  $\phi = 21^\circ, 110^\circ$  and  $\gamma = 0.5, 1, 2$ . In all the cases, the grid points are clustered near the duct walls, outlet, and inlet of the duct by using Eq. (25).

For validating the present results computed by applying the 3D WENOLBM solver, the outlet axial velocity profile for the six configurations at the Dean number  $K = 55$  considering the uniform velocity profile  $u_{ref} = 0.1$  at the inlet is compared with those presented by Hejranfar and Hajihassanpour<sup>26</sup> and Ghia and Sokhey,<sup>69</sup> which exhibit good agreement, as shown in Fig. 20.

For investigating the effect of the curvature and the aspect ratio on the flow characteristics, the sectional flow field and the velocity profile are shown in Figs. 21 and 22 for the curvature angles  $\phi = 21^\circ$  and  $\phi = 110^\circ$ , respectively, for three different values of the aspect ratio. The sectional velocity fields at the outlet of the duct are also shown in Fig. 23. It is observed in Fig. 23(b) that the center of the two recirculating regions at the duct outlet plane is placed nearly at the centerline of the outlet in the duct for the curvature angle  $\phi = 110^\circ$  for three aspect

TABLE V. Comparison of predicted separation angle and location of reattachment point for flow past the sphere with  $Re = 50, 100$ , and  $200$ .

| Re  | Author(s)                 | Method           | Reattachment point | $\theta_s$ (deg) |
|-----|---------------------------|------------------|--------------------|------------------|
| 50  | Taneda                    | Experiment       | 0.44               | ...              |
|     | Tomboulides and Orzag     | Spectral element | 0.405              | ...              |
|     | Jahnson and Patel         | N.S.             | 0.41               | 139.2            |
|     | Hejranfar and Ezzatneshan | CFDLBM           | 0.41               | 140.5            |
|     | Present study             | WENOLBM          | 0.41               | 140.1            |
| 100 | Taneda                    | Experiment       | 0.89               | ...              |
|     | Tomboulides               | Spectral element | 0.87               | 126.3            |
|     | Jahnson and Patel         | N.S.             | 0.88               | 126.5            |
|     | Nakamura                  | Experiment       | ...                | 126.3            |
|     | Hejranfar and Ezzatneshan | CFDLBM           | 0.87               | 127.8            |
|     | Present study             | WENOLBM          | 0.868              | 127.6            |
| 200 | Taneda                    | Experiment       | 1.161              | ...              |
|     | Tomboulides               | Spectral element | 1.433              | 115.81           |
|     | Nakamura                  | Experiment       | ...                | 116.24           |
|     | Present study             | WENOLBM          | 1.3725             | 113.70           |



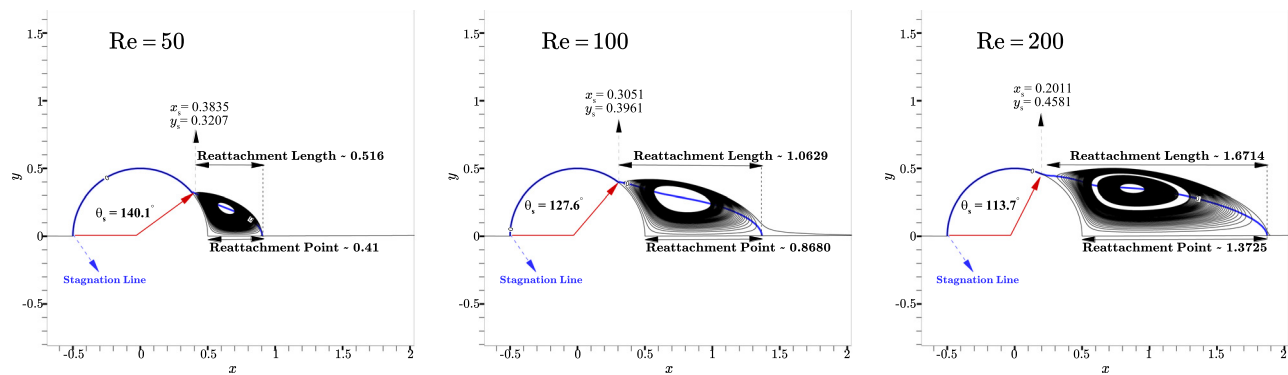


FIG. 32. Schematic illustration of wake length and separation angle for flow over the sphere with  $Re = 50, 100$ , and  $200$  (from left to right).

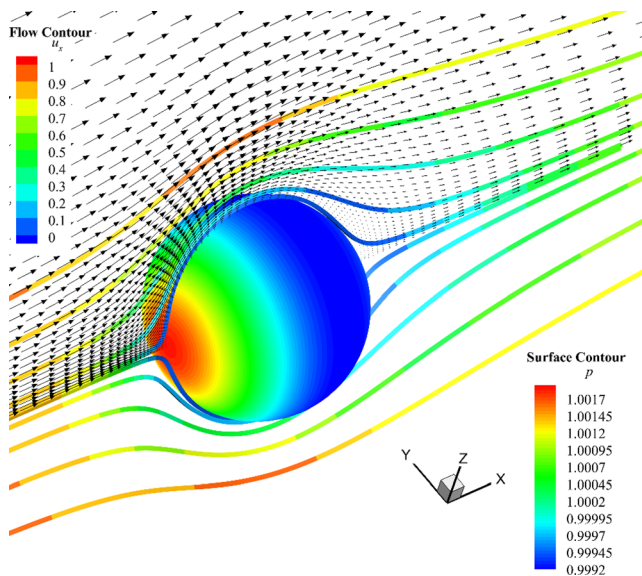


FIG. 33. Velocity field and surface pressure distribution for flow over the sphere with  $Re = 20$ .

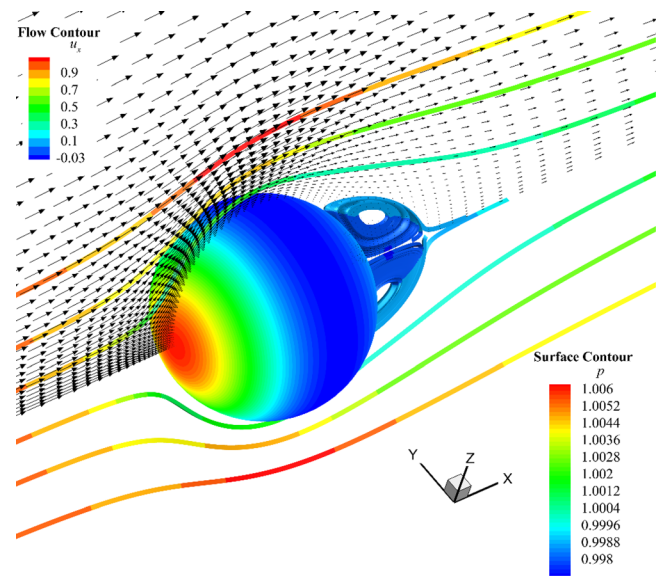


FIG. 34. Velocity field and surface pressure distribution for flow over the sphere with  $Re = 50$ .

ratios considered. Conversely, by increasing the value of the aspect ratio in the case with the curvature angle  $\varphi = 21^\circ$ , the center of the two recirculating regions is moved to the radial centerline from the duct outer wall, as shown in Fig. 23(a).

The study indicates that the existence of a non-unit aspect ratio makes the flow to be more complex and sensitive to the grid resolution. The grid refinement study shows that for achieving more accurate results for the non-unit aspect ratio cases, more number of the grid points are needed in comparison with the unit aspect ratio case. The convergence rate of the solution for the case with the curvature ratio  $\varphi = 110^\circ$  and the aspect ratio  $\gamma = 2$  is presented in Fig. 24 indicating that the steady-state solution is properly achieved. The simulation of this challenging internal flow problem verifies that the present solution methodology based on the 3D WENOLBM can be considered as an accurate and suitable solver for computing the 3D internal incompressible flows.

### E. Flow over the sphere

For examining the accuracy and capability of the present flow solver based on the 3D WENOLBM in simulating the external flows, the flow over a sphere is considered here as an appropriate test case owing the recirculating region occurring in the rear of this geometry. Numerous experimental and numerical investigations have been made about this problem for low Reynolds numbers,<sup>70–76</sup> and therefore, the present results can be assessed in comparison with these available results. Here, the Reynolds number  $Re = u_{ref}D/\nu$  is defined based on the free stream velocity  $u_{ref}$  and the sphere diameter  $D$ , in which the present simulation is performed by setting  $D = 1.0$  and the reference velocity is considered to be  $u_{ref} = 0.1$ , unless otherwise specified.

Taneda,<sup>70</sup> based on his experimental observations, reported the flow is completely laminar when  $Re < 22$  and the recirculating regions are not observed anywhere for this Reynolds number range. For

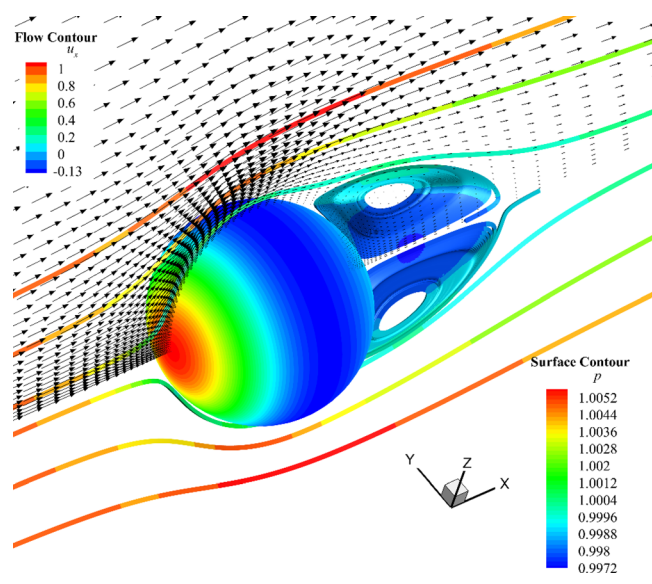


FIG. 35. Velocity field and surface pressure distribution for flow over the sphere with  $Re = 100$ .

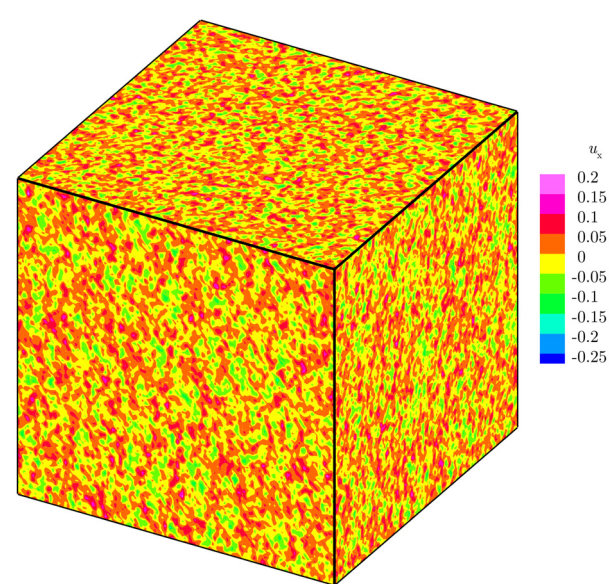


FIG. 37. Divergence-free initial velocity field using the procedure proposed by Kwak *et al.*

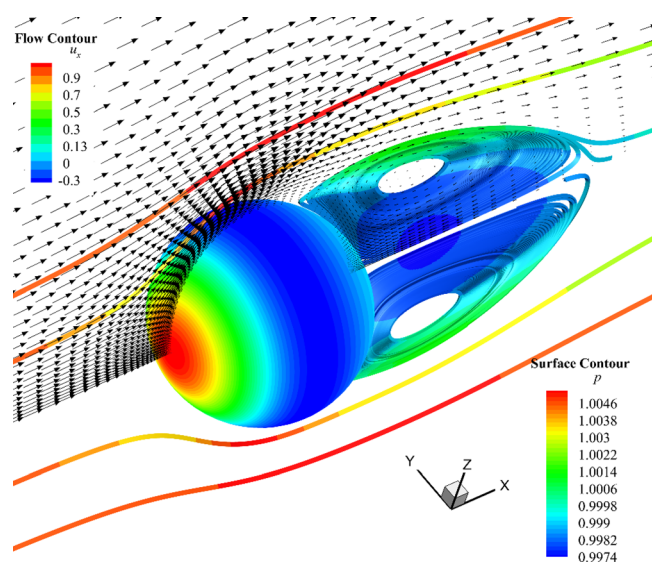


FIG. 36. Velocity field and surface pressure distribution for flow over the sphere with  $Re = 200$ .

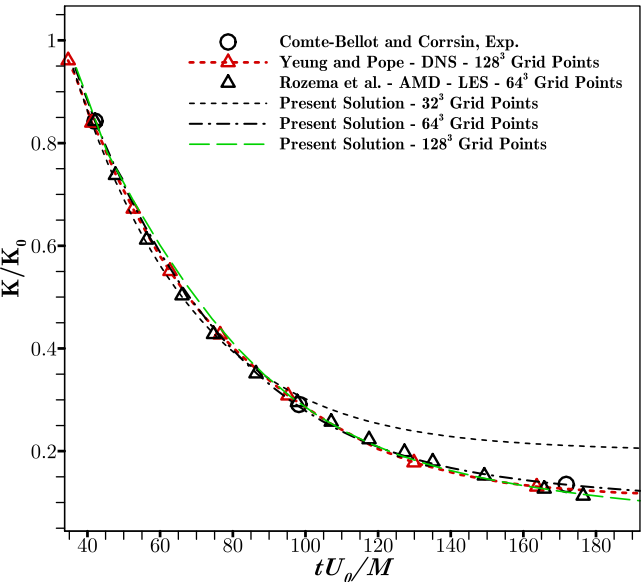


FIG. 38. Comparison of decay of turbulent kinetic energy in term of scaled time.

TABLE VI. Comparison of location of reattachment point for flow past the sphere with  $Re = 100$ .

| Author (s)                | Method                         | Computational domain                                  | Grid size                     | Reattachment point |
|---------------------------|--------------------------------|-------------------------------------------------------|-------------------------------|--------------------|
| Eitel-Amor <i>et al.</i>  | LBM with local mesh refinement | Cube with side sizes of $(64D \times 30D \times 30D)$ | $1.5 \times 10^6$ Cells       | 0.863              |
| Hejranfar and Ezzatneshan | CFDLBM                         | Sphere with radius of $50D$                           | $301 \times 10^3$ Grid points | 0.87               |
| Present study             | WENOLBM                        | Sphere with radius of $50D$                           | 262,144 Grid points           | 0.868              |

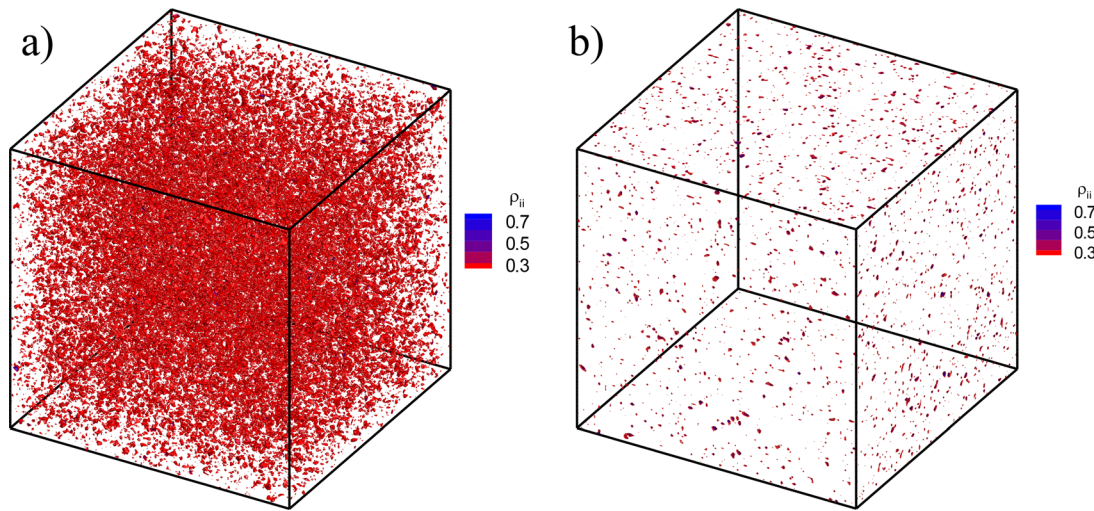


FIG. 39. Scaled iso-surfaces of Eulerian autocorrelation at (a)  $tU_0/M = 0$  and (b)  $tU_0/M = 180$ .

$Re \approx 22$ , a small circulation pattern is observed near the rear stagnation point and by increasing the  $Re$  number the recirculating region grows and a fully separated flow region is observed behind the sphere. He reported that the critical Reynolds number is about 24 and different values for the critical Reynolds number have been reported in other studies.

To examine the accuracy of the solution of the 3D WENOLBM in capturing the separated flow patterns due to the curved geometry, the flow is computed for four different Reynolds numbers  $Re = 20$ ,  $Re = 50$ ,  $Re = 100$ , and  $Re = 200$  and the corresponding results are compared with the available experimental and numerical results. Here, the grid system is clustered in the  $\xi$  direction near the stagnation points using Eq. (25) and near the sphere surface in the  $\eta$  direction using the flowing transformation available in Ref. 67

$$r_j = H \frac{(\beta + 1) - (\beta - 1)[(\beta + 1)/(\beta - 1)]^{1-\bar{\eta}_i}}{1 + [(\beta + 1)/(\beta - 1)]^{1-\bar{\eta}_i}}, \quad (25)$$

$$0 \leq \bar{\eta}_i = \frac{j-1}{j_{\max}-1} \leq 1,$$

where  $H$  is the distance between the sphere surface and the outer surface at the far-field boundary. Also, the radius of the O-mesh is considered 50-times larger than the sphere diameter. For removing the polar singularity on the stagnation lines, the grid points are generated with small distances far from these lines. The generated grid system with  $64 \times 64 \times 64$  grid points is shown in Fig. 25.

The pressure coefficient distribution over the sphere surface obtained by the 3D WENOLBM is compared with the numerical results presented by Hejranfar and Ezzatneshan,<sup>68</sup> and Tomboulides and Orszag<sup>71</sup> in Fig. 26 for  $Re = 50$ , 100, and 200. For further validating the present results, the surface pressure coefficient distribution is compared with the experimental results of Magnaudet *et al.*<sup>72</sup> and Le Clair *et al.*<sup>74</sup> for  $Re = 100$ , as shown in Fig. 27. The study shows that the present solution methodology provides reasonable results compared with the numerical and experiment ones. To further examine the accuracy of the solution of the 3D WENOLBM, the distribution of the surface pressure coefficient for the case  $Re = 20$  is given in Fig. 28

and compared with the semi-analytical solution carried out by Dennis and Walker<sup>77</sup> which show reasonable agreement.

The effect of the value of the reference velocity for the sphere test case is now investigated. For this aim, the steady-state solution of the flow over the sphere with  $Re = 200$  is obtained by applying the 3D WENOLBM with two values of the reference velocity,  $u_{\text{ref}} = 0.05$  and 0.1, considering the grid  $(64 \times 64 \times 64)$ . Figure 29 shows the corresponding surface pressure coefficient distributions, and it is indicated that the value of the reference velocity does not significantly affect the steady-state results of the incompressible flow over the sphere for  $u_{\text{ref}} \leq 0.1$ . To further check when the flow can be considered incompressible, one can calculate the divergence of the velocity vector ( $\nabla \cdot \mathbf{V}$ ) in the flow domain and the deviation from the criterion  $\nabla \cdot \mathbf{V} = 0$  indicates the level of the compressibility in the flow field. Figure 30 gives the distribution of the divergence of the velocity vector  $\nabla \cdot \mathbf{V}$  over the sphere surface for  $u_{\text{ref}} = 0.05$  and 0.1. The study shows that the compressibility effect decreases by reducing the value of the reference velocity and this level of the compressibility error does not affect the steady-state solution for  $u_{\text{ref}} \leq 0.1$ . Figure 31 illustrates the convergence history of the solution to reach the steady-state condition is considerably increased by decreasing the value of the reference velocity, and therefore, the corresponding computational time is significantly increased.

The separation angle and the wake length calculated in the present study are compared with the corresponding numerical and experimental results, as presented in Table V and schematically shown in Fig. 32 for  $Re$  numbers 50, 100, and 200. The results obtained by applying the 3D WENOLBM for these  $Re$  numbers are close to the available results. The velocity field and the surface pressure distribution for different  $Re$  numbers from 20 to 200 are shown in Figs. 33–36. The study shows that there is no circulation pattern behind the sphere at  $Re = 20$ . By increasing the  $Re$  number, a separated region is observed and the separation points symmetrically move far from the trailing edge of the sphere and the length of the flow wake increases. Table VI compares the length of the flow wake behind the sphere with those reported by Hejranfar and Ezzatneshan<sup>68</sup> and Eitel-Amor *et al.*<sup>76</sup> at  $Re = 100$  and good agreement is observed.



## F. Decay of homogeneous isotropic turbulence

To examine the accuracy and capability of the present three-dimensional high-order WENOLBM solver as the direct numerical simulation of the turbulent flows, a challenging benchmark test case, namely, the decay of the homogeneous isotropic turbulence, is numerically simulated here. The numerical setup for this test case is based on the decaying grid turbulence experiment performed by Comte-Bellot and Corrsin.<sup>78</sup> In their experiments, turbulence is generated by a grid with the mesh size  $M = 0.0508$  m for the flow with the mean velocity  $U_0 = 10$  m/s. Similar to Ref. 79, the present simulation is performed for the flow inside a cube of length  $L = 11M$  with considering the reference velocity  $u_{\text{ref}} = 0.02719 U_0$ .

To ensure of generating an isotropic incompressible initial velocity field, the simulation starts with a divergence-free velocity field, which is generated using the procedure prescribed by Kwak *et al.*<sup>80</sup> As a brief, this procedure begins with the establishment of an isotropic discrete velocity field  $\mathbf{u}(\mathbf{x})$  as follows:

$$\mathbf{u}(\mathbf{x}) = \sum_{n=-N/2}^{N/2-1} \sum \hat{\mathbf{u}}(\mathbf{k}_n) e^{i\mathbf{k}_n \cdot \mathbf{x}}, \quad (26)$$

where  $\hat{\mathbf{u}}(\mathbf{k}_n)$  is its discrete Fourier transform and the approximated energy spectrum is

$$E(k_n) \simeq 2\pi k_n^2 \langle \hat{\mathbf{u}}_i(\mathbf{k}_n) \hat{\mathbf{u}}_i^*(\mathbf{k}_n) \rangle, \quad (27)$$

where  $\mathbf{k}_n$  is the wave number vector defined by  $(k_n)_i = (2\pi/N\Delta)n_i$  in which  $n_i$  is an integer number between the nonsymmetrical interval of  $-N/2 \leq n_i \leq N/2 - 1$  for a grid system with  $N$  points in each direction. Here, the sign  $*$  donates a complex conjugate and  $\langle \rangle$  is considered as the volumetric average. Figure 37 shows the initial velocity field generated by this procedure for the grid system with  $128^3$  grid points. The implemented boundary conditions are considered periodic for all the boundary surfaces of the cube, which is confirmed by the initial velocity field.

To assess the accuracy and performance of the present 3D WENOLBM in resolving the structure of the homogenous isotropic turbulence, the temporal reduction of the turbulent kinetic energy, which is non-dimensionalized by the initial turbulent kinetic energy, in term of the scaled time  $\tilde{t} = tU_0/M$  is plotted in Fig. 38 and compared with the numerical results obtained by the direct numerical simulation performed by Yeung and Pope,<sup>81</sup> the anisotropic low dissipation (AMD) large eddy simulation by Rozema *et al.*<sup>79</sup> and the experimental data.<sup>80</sup> The comparison demonstrates excellent agreement between the present solution and the available results even when the coarse grids are used which indicates the suitability of the present WENOLBM for effectively computing such complicated flow structures. To obtain a better sensible sight of the reduction of the energy occurs in this flow, the iso-surfaces of the local distribution of the scaled Eulerian autocorrelation  $\rho_{ii} = u_i u_i / u_{\text{ref}}^2$  are shown in Fig. 39 for the initial condition  $\tilde{t} = tU_0/M = 0$  when the non-dimensional turbulent kinetic energy has its maximum amount and at the scaled time  $\tilde{t} = tU_0/M = 180$  that the flow structures are almost decayed.

The study shows that the present three-dimensional high-order WENOLBM solver can be considered as an appropriate candidate for the direct numerical simulation of the incompressible turbulent flows. Here, a geometrically simple test case is simulated to examine this

capability. The present solution method implemented on the generalized curvilinear coordinates can handle similar simulations over more complicated geometries by an appropriate parallelization on central processing unit (CPU)/graphics processing unit (GPU) to be performed on the current flow solver.

## VII. CONCLUSION

In the present work, the high-order WENO finite-difference LBM (WENOLBM) is extended and applied in the three-dimensional generalized curvilinear coordinates for effectively computing the three-dimensional incompressible flow problems. In addition to the high-order accuracy solutions provided by the numerical solution procedure adopted here, it does not need any filtering technique for the solution to be stable. Different internal and external incompressible flow problems with and without curved boundaries are computed to assess the accuracy and robustness of the solution methodology presented. It is shown that the results obtained by applying the three-dimensional WENOLBM for the test cases simulated are in agreement with the available numerical and experimental results. The introduced solution procedure can be considered as an accurate and suitable computational technique for studying physical phenomena and simulating fluid dynamics over practical and realistic geometries. Such high-order accurate LB-based flow solver by implementing appropriate parallelization on CPU/GPU can be considered as a good candidate to be used for the direct numerical simulations of the incompressible flows.

## ACKNOWLEDGMENTS

The authors would like to thank Sharif University of Technology for the support of this research.

## AUTHOR DECLARATIONS

### Conflict of Interest

The authors have no conflicts to disclose.

### Author Contributions

**Kazem Hejranfar:** Conceptualization (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Supervision (equal); Writing – review & editing (equal). **Mohammad Abotalebi:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Validation (equal); Visualization (equal); Writing – original draft (equal).

## DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

## REFERENCES

- <sup>1</sup>F. Nannelli and S. Succi, "The lattice Boltzmann equation on irregular lattices," *J. Stat. Phys.* **68**, 401–407 (1992).
- <sup>2</sup>R. Mei and W. Shyy, "On the finite difference-based lattice Boltzmann method in curvilinear coordinates," *J. Comput. Phys.* **143**(2), 426–448 (1998).
- <sup>3</sup>X. He and L.-S. Luo, "Theory of the lattice Boltzmann method: From the Boltzmann equation to the lattice Boltzmann equation," *Phys. Rev. E* **56**(6), 6811 (1997).

- <sup>4</sup>H. Xi, G. Peng, and S.-H. Chou, "Finite-volume lattice Boltzmann method," *Phys. Rev. E* **59**(5), 6202 (1999).
- <sup>5</sup>M. Stiebler, J. Tölke, and M. Krafczyk, "An upwind discretization scheme for the finite volume lattice Boltzmann method," *Comput. Fluids* **35**(8–9), 814–819 (2006).
- <sup>6</sup>K. Qu, C. Shu, and Y. T. Chew, "Simulation of shock-wave propagation with finite volume lattice Boltzmann method," *Int. J. Mod. Phys. C* **18**(04), 447–454 (2007).
- <sup>7</sup>D. V. Patil and K. Lakshmisha, "Finite volume TVD formulation of lattice Boltzmann simulation on unstructured mesh," *J. Comput. Phys.* **228**(14), 5262–5279 (2009).
- <sup>8</sup>A. Zarghami, C. Biscarini, S. Succi, and S. Ubertini, "Hydrodynamics in porous media: A finite volume lattice Boltzmann study," *J. Sci. Comput.* **59**, 80–103 (2014).
- <sup>9</sup>W. Li and L.-S. Luo, "Finite volume lattice Boltzmann method for nearly incompressible flows on arbitrary unstructured meshes," *Commun. Comput. Phys.* **20**(2), 301–324 (2016).
- <sup>10</sup>Y. Wang *et al.*, "A simplified finite volume lattice Boltzmann method for simulations of fluid flows from laminar to turbulent regime, Part I: Numerical framework and its application to laminar flow simulation," *Comput. Math. Appl.* **79**(5), 1590–1618 (2020).
- <sup>11</sup>K. Hejranfar, H. H. Nasab, and M. H. Azampour, "Arbitrary Lagrangian–Eulerian unstructured finite-volume lattice-Boltzmann method for computing two-dimensional compressible inviscid flows over moving bodies," *Phys. Rev. E* **101**(2), 023308 (2020).
- <sup>12</sup>M. Junk, "A finite difference interpretation of the lattice Boltzmann method," *Numer. Methods Partial Differ. Equations* **17**(4), 383–402 (2001).
- <sup>13</sup>G. Kefayati, "Implementing vorticity–velocity formulation in a finite difference lattice Boltzmann method for two-dimensional incompressible generalized Newtonian fluids," *Phys. Fluids* **36**(1), 013128 (2024).
- <sup>14</sup>G. Kefayati, A. Tolouiyani, and A. P. Dyson, "Finite difference lattice Boltzmann method for modeling dam break debris flows," *Phys. Fluids* **35**(1), 013102 (2023).
- <sup>15</sup>A. Fakhari and T. Lee, "Numerics of the lattice Boltzmann method on nonuniform grids: Standard LBM and finite-difference LBM," *Comput. Fluids* **107**, 205–213 (2015).
- <sup>16</sup>K. Hejranfar and E. Ezzatneshan, "A high-order compact finite-difference lattice Boltzmann method for simulation of steady and unsteady incompressible flows," *Int. J. Numer. Methods Fluids* **75**(10), 713–746 (2014).
- <sup>17</sup>K. Hejranfar and E. Ezzatneshan, "Implementation of a high-order compact finite-difference lattice Boltzmann method in generalized curvilinear coordinates," *J. Comput. Phys.* **267**, 28–49 (2014).
- <sup>18</sup>Y. Sun and Z. Tian, "High-order upwind compact finite-difference lattice Boltzmann method for viscous incompressible flows," *Comput. Math. Appl.* **80**(7), 1858–1872 (2020).
- <sup>19</sup>C. Nor Azwadi and T. Tanahashi, "Simplified finite difference thermal lattice Boltzmann method," *Int. J. Mod. Phys. B* **22**(22), 3865–3876 (2008).
- <sup>20</sup>Z. Guo and T.-S. Zhao, "Explicit finite-difference lattice Boltzmann method for curvilinear coordinates," *Phys. Rev. E* **67**(6), 066709 (2003).
- <sup>21</sup>K. Hejranfar, M. H. Saadat, and S. Taheri, "High-order weighted essentially nonoscillatory finite-difference formulation of the lattice Boltzmann method in generalized curvilinear coordinates," *Phys. Rev. E* **95**(2), 023314 (2017).
- <sup>22</sup>K. Hejranfar and M. H. Saadat, "Preconditioned WENO finite-difference lattice Boltzmann method for simulation of incompressible turbulent flows," *Comput. Math. Appl.* **76**(6), 1427–1446 (2018).
- <sup>23</sup>X. Shi, J. Lin, and Z. Yu, "Discontinuous Galerkin spectral element lattice Boltzmann method on triangular element," *Int. J. Numer. Methods Fluids* **42**(11), 1249–1261 (2003).
- <sup>24</sup>M. Min and T. Lee, "A spectral-element discontinuous Galerkin lattice Boltzmann method for nearly incompressible flows," *J. Comput. Phys.* **230**(1), 245–259 (2011).
- <sup>25</sup>Y. Li, E. J. LeBoeuf, and P. Basu, "Least-squares finite-element lattice Boltzmann method," *Phys. Rev. E* **69**(6), 065701 (2004).
- <sup>26</sup>K. Hejranfar and M. Hajihassanpour, "Chebyshev collocation spectral lattice Boltzmann method in generalized curvilinear coordinates," *Comput. Fluids* **146**, 154–173 (2017).
- <sup>27</sup>K. Hejranfar and M. Hajihassanpour, "Chebyshev collocation spectral lattice Boltzmann method for simulation of low-speed flows," *Phys. Rev. E* **91**(1), 013301 (2015).
- <sup>28</sup>K. Hejranfar and A. Ghaffarian, "A spectral difference lattice Boltzmann method for solution of inviscid compressible flows on structured grids," *Comput. Math. Appl.* **72**(5), 1341–1368 (2016).
- <sup>29</sup>K. Hejranfar and A. Ghaffarian, "A high-order accurate unstructured spectral difference lattice Boltzmann method for computing inviscid and viscous compressible flows," *Aerosp. Sci. Technol.* **98**, 105661 (2020).
- <sup>30</sup>D. d'Humières, "Multiple-relaxation-time lattice Boltzmann models in three dimensions," *Philos. Trans. R. Soc., A* **360**(1792), 437–451 (2002).
- <sup>31</sup>R. Du, B. Shi, and X. Chen, "Multi-relaxation-time lattice Boltzmann model for incompressible flow," *Phys. Lett. A* **359**(6), 564–572 (2006).
- <sup>32</sup>M. Geier, A. Greiner, and J. G. Korvink, "Cascaded digital lattice Boltzmann automata for high Reynolds number flow," *Phys. Rev. E* **73**(6), 066705 (2006).
- <sup>33</sup>P. Asinari, "Generalized local equilibrium in the cascaded lattice Boltzmann method," *Phys. Rev. E* **78**(1), 016701 (2008).
- <sup>34</sup>F. Bösch, S. S. Chikatamarla, and I. V. Karlin, "Entropic multirelaxation lattice Boltzmann models for turbulent flows," *Phys. Rev. E* **92**(4), 043309 (2015).
- <sup>35</sup>S. Ansumali and I. V. Karlin, "Single relaxation time model for entropic lattice Boltzmann methods," *Phys. Rev. E* **65**(5), 056312 (2002).
- <sup>36</sup>E. Xing *et al.*, "Numerical investigation of flow past a cylinder using cumulant lattice Boltzmann method," *Phys. Fluids* **36**(3), 035166 (2024).
- <sup>37</sup>M. Geier, M. Schönherr, A. Pasquali, and M. Krafczyk, "The cumulant lattice Boltzmann equation in three dimensions: Theory and validation," *Comput. Math. Appl.* **70**(4), 507–547 (2015).
- <sup>38</sup>X.-D. Liu, S. Osher, and T. Chan, "Weighted essentially non-oscillatory schemes," *J. Comput. Phys.* **115**(1), 200–212 (1994).
- <sup>39</sup>G.-S. Jiang and C.-W. Shu, "Efficient implementation of weighted ENO schemes," *J. Comput. Phys.* **126**(1), 202–228 (1996).
- <sup>40</sup>C.-W. Shu, "Essentially non-oscillatory and weighted essentially non-oscillatory schemes," *Acta Numerica* **29**, 701–762 (2020).
- <sup>41</sup>D. S. Balsara and C.-W. Shu, "Monotonicity preserving weighted essentially non-oscillatory schemes with increasingly high order of accuracy," *J. Comput. Phys.* **160**(2), 405–452 (2000).
- <sup>42</sup>S. Pirozzoli, "Conservative hybrid compact-WENO schemes for shock-turbulence interaction," *J. Comput. Phys.* **178**(1), 81–117 (2002).
- <sup>43</sup>S. Gottlieb, J. S. Mullen, and S. J. Ruuth, "A fifth order flux implicit WENO method," *J. Sci. Comput.* **27**(1), 271–287 (2006).
- <sup>44</sup>H. M. Darian and V. Esfahanian, "Assessment of WENO schemes for multi-dimensional Euler equations using GPU," *Int. J. Numer. Methods Fluids* **76**(12), 961–981 (2014).
- <sup>45</sup>G.-S. Jiang and C.-c. Wu, "A high-order WENO finite difference scheme for the equations of ideal magnetohydrodynamics," *J. Comput. Phys.* **150**(2), 561–594 (1999).
- <sup>46</sup>J. Kleimann *et al.*, "Three-dimensional MHD high-resolution computations with CWENO employing adaptive mesh refinement," *Comput. Phys. Commun.* **158**(1), 47–56 (2004).
- <sup>47</sup>A. J. Christlieb, J. A. Rossmann, and Q. Tang, "Finite difference weighted essentially non-oscillatory schemes with constrained transport for ideal magnetohydrodynamics," *J. Comput. Phys.* **268**, 302–325 (2014).
- <sup>48</sup>A. J. Christlieb, X. Feng, Y. Jiang, and Q. Tang, "A high-order finite difference WENO scheme for ideal magnetohydrodynamics on curvilinear meshes," *SIAM J. Sci. Comput.* **40**(4), A2631–A2666 (2018).
- <sup>49</sup>E. Johnsen and T. Colonius, "Implementation of WENO schemes in compressible multicomponent flow problems," *J. Comput. Phys.* **219**(2), 715–732 (2006).
- <sup>50</sup>R. Borges, M. Carmona, B. Costa, and W. S. Don, "An improved weighted essentially non-oscillatory scheme for hyperbolic conservation laws," *J. Comput. Phys.* **227**(6), 3191–3211 (2008).
- <sup>51</sup>Y. Ha, C. H. Kim, Y. J. Lee, and J. Yoon, "An improved weighted essentially non-oscillatory scheme with a new smoothness indicator," *J. Comput. Phys.* **232**(1), 68–86 (2013).
- <sup>52</sup>R. Abedian, H. Adibi, and M. Dehghan, "A high-order weighted essentially non-oscillatory (WENO) finite difference scheme for nonlinear degenerate parabolic equations," *Comput. Phys. Commun.* **184**(8), 1874–1888 (2013).
- <sup>53</sup>K. Hejranfar and S. Rahmani, "Numerical simulation of shock-disturbances interaction in high-speed compressible inviscid flow over a blunt nose using weighted essentially non-oscillatory scheme," *Wave Motion* **88**, 167–195 (2019).

- <sup>54</sup>S. Rahmani and K. Hejranfar, "Numerical study of shock-disturbances interaction in hypersonic inviscid flows with real gas effects using high-order WENO scheme," *Comput. Fluids* **229**, 105079 (2021).
- <sup>55</sup>J.-Y. Yang, S.-C. Yang, Y.-N. Chen, and C.-A. Hsu, "Implicit weighted ENO schemes for the three-dimensional incompressible Navier-Stokes equations," *J. Comput. Phys.* **146**(1), 464–487 (1998).
- <sup>56</sup>J. Zhang and T. L. Jackson, "A high-order incompressible flow solver with WENO," *J. Comput. Phys.* **228**(7), 2426–2442 (2009).
- <sup>57</sup>J.-M. Qiu and C.-W. Shu, "Conservative high order semi-Lagrangian finite difference WENO methods for advection in incompressible flow," *J. Comput. Phys.* **230**(4), 863–889 (2011).
- <sup>58</sup>F. Meng, J. Banks, W. Henshaw, and D. Schwendeman, "Fourth-order accurate fractional-step IMEX schemes for the incompressible Navier-Stokes equations on moving overlapping grids," *Comput. Methods Appl. Mech. Eng.* **366**, 113040 (2020).
- <sup>59</sup>P. L. Bhatnagar, E. P. Gross, and M. Krook, "A model for collision processes in gases. I. Small amplitude processes in charged and neutral one-component systems," *Phys. Rev.* **94**(3), 511 (1954).
- <sup>60</sup>T. Krüger *et al.*, "The lattice Boltzmann equation," in *The Lattice Boltzmann Method: Principles and Practice* (Springer, Cham, 2017), pp. 61–104.
- <sup>61</sup>X. He and L.-S. Luo, "Lattice Boltzmann model for the incompressible Navier-Stokes equation," *J. Stat. Phys.* **88**, 927–944 (1997).
- <sup>62</sup>A. Jameson, "Numerical solution of the Euler equations for compressible inviscid fluids," Report No. MAE-1643, 1985.
- <sup>63</sup>D. I. Ketcheson, "Highly efficient strong stability-preserving Runge-Kutta methods with low-storage implementations," *SIAM J. Sci. Comput.* **30**(4), 2113–2136 (2008).
- <sup>64</sup>X. He, G. D. Doolen, and T. Clark, "Comparison of the lattice Boltzmann method and the artificial compressibility method for Navier-Stokes equations," *J. Comput. Phys.* **179**(2), 439–451 (2002).
- <sup>65</sup>U. Ghia, K. N. Ghia, and C. Shin, "High-Re solutions for incompressible flow using the Navier-Stokes equations and a multigrid method," *J. Comput. Phys.* **48**(3), 387–411 (1982).
- <sup>66</sup>C. R. Ethier and D. Steinman, "Exact fully 3D Navier-Stokes solutions for benchmarking," *Int. J. Numer. Methods Fluids* **19**(5), 369–375 (1994).
- <sup>67</sup>K. A. Hoffmann and S. T. Chiang, *Computational Fluid Dynamics Volume 1* (Engineering Education System, 2000).
- <sup>68</sup>E. Ezzatneshan and K. Hejranfar, "Simulation of three-dimensional incompressible flows in generalized curvilinear coordinates using a high-order compact finite-difference lattice Boltzmann method," *Int. J. Numer. Methods Fluids* **89**(7), 235–255 (2019).
- <sup>69</sup>K. Ghia and J. Sokhey, "Laminar incompressible viscous flow in curved ducts of regular cross-sections," *J. Fluids Eng.* **99**, 640–648 (1977).
- <sup>70</sup>S. Taneda, "Experimental investigation of the wake behind a sphere at low Reynolds numbers," *J. Phys. Soc. Jpn.* **11**(10), 1104–1108 (1956).
- <sup>71</sup>A. G. Tomboulides and S. A. Orszag, "Numerical investigation of transitional and weak turbulent flow past a sphere," *J. Fluid Mech.* **416**, 45–73 (2000).
- <sup>72</sup>J. Magnaudet, M. Rivero, and J. Fabre, "Accelerated flows past a rigid sphere or a spherical bubble. Part 1. Steady straining flow," *J. Fluid Mech.* **284**, 97–135 (1995).
- <sup>73</sup>T. Johnson and V. Patel, "Flow past a sphere up to a Reynolds number of 300," *J. Fluid Mech.* **378**, 19–70 (1999).
- <sup>74</sup>L. Clair, B. A. Hamielec, and H. Pruppacher, "A numerical study of the drag on a sphere at low and intermediate Reynolds numbers," *J. Atmos. Sci.* **27**(2), 308–315 (1970).
- <sup>75</sup>I. Nakamura, "Steady wake behind a sphere," *Phys. Fluids* **19**(1), 5–8 (1976).
- <sup>76</sup>G. Eitel-Amor, M. Meinke, and W. Schröder, "A lattice-Boltzmann method with hierarchically refined meshes," *Comput. Fluids* **75**, 127–139 (2013).
- <sup>77</sup>S. Dennis and J. Walker, "Calculation of the steady flow past a sphere at low and moderate Reynolds numbers," *J. Fluid Mech.* **48**(4), 771–789 (1971).
- <sup>78</sup>G. Comte-Bellot and S. Corrsin, "Simple Eulerian time correlation of full-and narrow-band velocity signals in grid-generated, 'isotropic' turbulence," *J. Fluid Mech.* **48**(2), 273–337 (1971).
- <sup>79</sup>W. Rozema, H. J. Bae, P. Moin, and R. Verstappen, "Minimum-dissipation models for large-eddy simulation," *Phys. Fluids* **27**(8), 085107 (2015).
- <sup>80</sup>D. Kwak, W. C. Reynolds, and J. H. Ferziger, "Three-dimensional time dependent computation of turbulent flow," Report No. TF-5, 1975.
- <sup>81</sup>P.-K. Yeung and S. B. Pope, "Lagrangian statistics from direct numerical simulations of isotropic turbulence," *J. Fluid Mech.* **207**, 531–586 (1989).